

CHEM 396: Simulation of Amorphous Silicon using MLIPs

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Abstract

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1 Introduction

Amorphous silicon (a-Si) lacks the long-range order of a crystal which its crystalline counterpart possesses. Most atoms are still bonded to four neighbours in roughly tetrahedral arrangements, but the bond lengths and angles vary rather than being fixed, a structure described by the continuous random network model. a-Si is used in multiple applications, and its properties depend strongly on coordination defects, where atoms have too few or too many bonds.

Simulating a-Si requires a model of the forces between atoms, which involves a tradeoff. Density functional theory (DFT) is accurate but computationally expensive, while empirical potentials¹ are fast but less flexible. Machine-learning potentials offer a middle ground by learning the energy from DFT data. This project uses one such method, the Moment Tensor Potential (MTP).

The aim of this project is to generate models of a-Si through melt-quench molecular dynamics² simulations driven by an MTP, and to analyze the resulting structures using the Morphological Autoregressive Protocol (MAP).

2 Background

2.1 Amorphous Silicon

2.1.1 Silicon

Definition 1 (Silicon). *Silicon is a chemical element with atomic number 14, symbol Si. It is a metalloid with properties between metals and nonmetals, and is the second most abundant element in the Earth’s crust after oxygen.*

Definition 2 (Crystalline Silicon). *Crystalline silicon (c-Si) is a solid form of silicon where atoms are arranged in a regular, repeating lattice structure, specifically the diamond cubic structure.*

Definition 3 (Amorphous Silicon). *Amorphous silicon (a-Si) is a form of silicon that lacks long-range crystalline order, exhibiting a disordered atomic structure.*

¹Empirical potentials are mathematical models that describe the interaction between atoms based on experimental data.

²You heat simulated silicon until it’s liquid, then cool it fast so it freezes into a disordered solid.

	c-Si	a-Si
Hybridization	sp^3	sp^3
Bulk Modulus (GPa) ³	98 ⁴	70–80
Folds	4-fold	4-fold with coordination defects (5)
Density (g/cm ³) ⁵	2.33	2.28
Bond Length (Å)	2.35	~2.35
Bond Angle (°)	109.5	109.5 ± 10

Table 1: Comparison of crystalline and amorphous silicon.

2.1.2 The Continuous Random Network Model

Definition 4 (Continuous Random Network). *The continuous random network (CRN) model describes an amorphous material as a non-crystalline 3D network with no repeating long-range structure.*

³The material’s resistance to uniform compression or how much its volume decreases when subjected to hydrostatic pressure. In this context, at room temperature and atmospheric pressure.

⁴This structure gives c-Si its characteristic properties, such as high thermal conductivity and semiconducting behaviour.

⁵At room temperature.

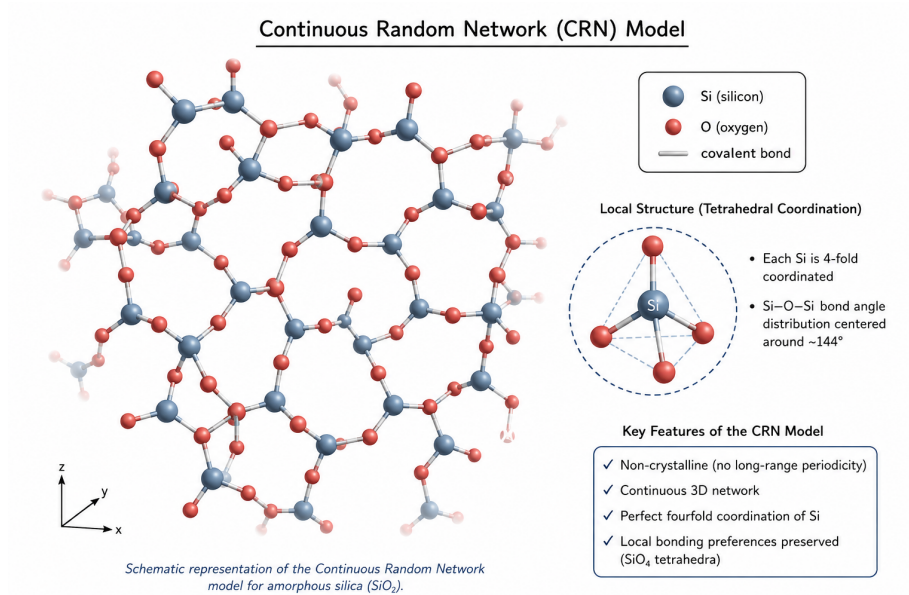


Figure 1: The continuous random network model of amorphous silica.

A CRN is disordered by construction. Because the network is non-periodic, bond lengths stay tightly clustered near the crystalline value, bond angles spread around the tetrahedral value of $109.47^\circ \pm 10^\circ$, and the dihedral angles⁶ become essentially random. This is the key difference from periodic c-Si, where all three take sharp fixed values.

The geometric freedom in such a network is limited. For an amorphous solid to form a stable, fully connected network in the first place, its local bonding must satisfy a small set of structural constraints, first set out by Zachariasen [9].

Theorem 1 (Zachariasen’s Rules for Glass Formation).

- *No oxygen atom may be linked to more than two cations.*
- *The cation coordination number is small⁷.*
- *Oxygen polyhedra⁸ must share corners, not edges or faces.*
- *For 3D networks, at least three corners must be shared by polyhedra.*

Zachariasen’s rules describe what a valid glass network must look like, but they tell us which structures are admissible without prescribing how to build one. To actually generate an atomic model that satisfies these constraints, we

⁶The dihedral angle of a chain of four bonded atoms is the angle between the plane of the first three and the plane of the last three.

⁷For example, 3 for Boron and 4 for Silicon

⁸A three-dimensional solid with flat faces and straight edges.

need an explicit procedure, which is precisely what the Wooten-Winer-Weaire algorithm provides [10].

Theorem 2 (Wooten-Winer-Weaire Algorithm). *The Wooten-Winer-Weaire (WWW) algorithm is a foundational Monte Carlo method used to generate realistic, CRN models of amorphous tetrahedral semiconductors.*

At the core of the algorithm is a local operation called a bond transposition. In this step, two nearby bonds are broken and the atoms are reconnected in a different way. This changes the local structure of the network while keeping each atom bonded to four neighbors. The new structure is then accepted or rejected using a Monte Carlo method based on how the strain energy changes. Fig. 12 illustrates this process step by step.

2.1.3 Coordination Defects

Definition 5 (Coordination Defect). *A coordination defect is an atom that deviates from the ideal coordination (fourfold for silicon), such as a dangling bond (threefold) or a floating bond (fivefold).*

Definition 6 (Dangling Bond). *A dangling bond is an under-coordinated (threefold or less) silicon atom left with an unpaired electron. The site is chemically reactive and carries a local magnetic moment, making it visible in electron spin resonance (ESR).*

Definition 7 (Floating Bond). *A floating bond is an over-coordinated (fivefold or more) silicon atom. Its excess electron delocalizes over the neighbouring bonds rather than localizing, so unlike a dangling bond it carries no strong magnetic moment and is ESR-silent.*

Remark 1. *The concentration of coordination defects is not intrinsic to a-Si but depends strongly on sample preparation.*

Experimentally, as-implanted a-Si is highly defective and strained, but these defects anneal out under thermal relaxation as the network relaxes toward a lower-energy state. The same sensitivity appears in simulation: melt-quench MD traps coordination defects, increasingly so at faster quench⁹ rates, whereas WWW bond-switching relaxes toward near-perfect fourfold coordination. Reverse Monte Carlo (RMC), the activation-relaxation technique (ARTn), and force-enhanced atomic refinement (FEAR) each strike a different balance between matching experimental data and minimizing defect density.

Definition 8 (Reverse Monte Carlo). *Reverse Monte Carlo (RMC) randomly moves atoms until the model’s computed $g(r)$ and structure factor match experimental data. It fits the measurement rather than minimizing energy, so different structures can give the same fit.*

⁹Quench is rapid cooling.

2.1.4 Ring Defects and Medium-Range Order

Definition 9 (Ring in a Network). *A ring is a closed loop of bonds connecting atoms.*

Definition 10 (Ring Defect). *Ring defects describe how these loops vary in size. In a-Si, many rings are not six-membered. These distortions in ring size create local strain, change bond angles, and can introduce electronic gap states.*

Definition 11 (Medium-Range Order). *Medium-range order (MRO) is structure that persists over a few atomic spacings, beyond nearest neighbours but short of the long-range periodicity of a crystal.*

In realistic a-Si models the ring distribution peaks around five-, six-, and seven-membered rings. Three-membered rings are essentially absent, and four-membered rings are rare because both are heavily strained¹⁰.

One practical caveat is that there is no single agreed-upon rule for what counts as a “ring” in a disordered network. Several definitions exist (King or shortest-path, Guttman, and primitive rings), and each can count the loops in the same structure differently.

Definition 12 (King Criterion). *A ring-counting rule: for each pair of bonds at an atom, the ring is the shortest path connecting them back through the network.*

2.1.5 The Paracrystallinity Debate

A long-standing debate concerns whether a-Si is genuinely a pure continuous random network, or whether real samples contain small ordered regions mixed into the disordered matrix.

Definition 13 (Paracrystalline Model). *A model in which a-Si contains small crystal-like regions of local order (“paracrystallites”) embedded in a disordered matrix, so the structure retains correlations at the medium range rather than being fully random.*

Support for the paracrystalline picture comes from fluctuation electron microscopy (FEM), a technique that can pick up medium-range order which ordinary diffraction tends to average away. Treacy and Borisenko (2012) [11] read their FEM signals as evidence for tiny paracrystalline regions. This reading was questioned by others, and the issue stayed unresolved because the measurement is indirect rather than a direct image of the structure. More recently, the debate has reopened from the simulation side, Rosset, Drabold and Deringer (2025) [7] find paracrystalline order appearing in large machine-learning MD models of a-Si, suggesting that the pure CRN may not fully describe real samples.

¹⁰The bonds are forced into shapes they don’t want to be in, so they store energy and are unstable, like a bent spring that wants to snap back.

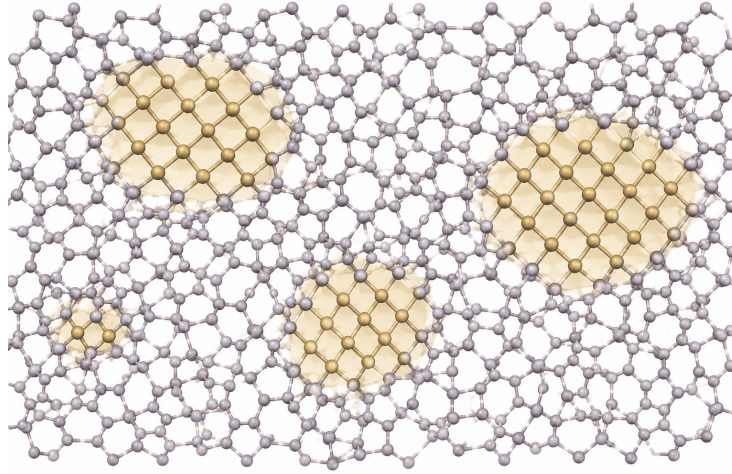


Figure 2: Schematic illustration of the paracrystalline model of a-Si. The highlighted ordered domains are illustrative and are not drawn to scale.

2.1.6 Applications

a-Si is used predominantly as a semiconductor for large-area and flexible electronics. Its major commercial and industrial applications include:

- **Active-matrix displays.** a-Si is the foundational material for the thin-film transistor (TFT) arrays in liquid-crystal displays (LCDs), controlling individual pixels in laptops and televisions.
- **Photovoltaics and solar power.** It is used in lightweight, flexible, and semi-transparent thin-film solar panels. Although less efficient than c-Si, a-Si panels perform well in low-light conditions.
- **Medical X-ray imaging.** In large-area digital X-ray flat-panel detectors, a-Si arrays convert X-rays into high-resolution digital images.
- **Sensors and scanners.** a-Si is deployed in document scanners, electronic whiteboards, and optical colour sensors because it can cover large surface areas.
- **Gravitational wave detectors** Like LIGO¹¹ use a-Si coatings on mirrors to reduce thermal noise and improve sensitivity.
- **Quantum computers** Use a-Si in superconducting qubits and Josephson junctions¹², where its low defect density and tunable properties help maintain coherence.

¹¹Laser Interferometer Gravitational-Wave Observatory.

¹²Josephson junctions are weak links between two superconductors that allow for the flow of Cooper pairs. PS: uses amorphous silicon as its weak-link or tunneling barrier material.

2.2 Molecular Dynamics

2.2.1 Equations of Motion

Definition 14 (Molecular Dynamics). *Molecular dynamics (MD) is a computational method for simulating the physical movements of atoms and molecules.*

Molecular dynamics evolves a system of atoms in time by integrating Newton’s equations of motion. This requires the forces on each atom at every timestep, which means a model of the potential energy as a function of atomic positions. That model is the interatomic potential.

Definition 15 (Force Field). *A force field¹³ is the complete set of functions and fitted parameters used to compute the energy and forces of a system.*

Remark 2. *Potentials are classified by where their parameters come from.*

Definition 16 (*Ab initio*). *Energy computed directly from quantum mechanics, no parameters fitted to data. DFT is the standard example. Accurate but expensive.*

Definition 17 (Empirical and semi-empirical). *A fixed functional form chosen on physical grounds, with parameters fitted to experimental or ab initio data. Lennard-Jones and Stillinger-Weber are of this kind. Cheap, but limited by the rigidity of the chosen form.*

Definition 18 (Machine-learned). *No fixed functional form. The structure-energy relationship is learned from a large set of ab initio reference calculations. MTPs belong here.*

2.2.2 Thermodynamic Ensembles

Definition 19 (Thermodynamic Ensemble). *The set of quantities held fixed during a simulation.*

Three ensembles are used in this project.

Definition 20 (NVE, microcanonical). *Constant number of atoms, volume, and total energy. Newton’s equations are integrated with no thermostat, so the system exchanges nothing with its surroundings.*

Definition 21 (NVT, canonical). *Constant number of atoms, volume, and temperature. A thermostat exchanges energy with the system to hold the temperature at a target value while the box stays fixed.*

Definition 22 (NPT, isothermal-isobaric). *Constant number of atoms, pressure, and temperature. A barostat is added to the thermostat, letting the box expand or contract until the system reaches the target pressure. Since the volume is free to change, this is the ensemble used to find the density a potential actually prefers.*

¹³“Force field” and “interatomic potential” are used interchangeably, the former more common in chemistry, the latter in materials physics.

2.3 Interatomic Potentials

Definition 23 (Interatomic Potential). *An interatomic potential is a mathematical function that describes the potential energy of a system of atoms as a function of their positions. It is used to calculate forces and energies in MD simulations.*

2.3.1 Lennard-Jones Potential

Definition 24 (Lennard-Jones Potential). *The Lennard-Jones (LJ) potential is a total interaction energy combining an attractive and a repulsive term. The attraction comes from London dispersion forces (instantaneous induced-dipole-induced-dipole interactions), and the repulsion arises from Pauli exclusion when the electron clouds of two atoms overlap at short range.*

The LJ potential is the simplest widely used empirical pair potential:

$$V_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (1)$$

It has only two free parameters. The distance σ is the separation r at which the potential energy crosses zero, that is, where attraction and repulsion exactly cancel. Physically, σ acts as the effective diameter of an atom. Two atoms strongly repel once they come closer than about σ . The lowest-energy separation (the bottom of the well, where the atoms sit at rest) is slightly larger, at $r = 2^{1/6}\sigma$.

The energy ϵ is the depth of the potential well, how much energy you would have to supply to pull two bonded atoms apart to infinity, which sets how strongly two neighbouring atoms are bound.

Changing σ and ϵ only rescales the curve along the r and V axes respectively; the overall shape stays the same regardless of how they are tuned.

Definition 25 (Lennard-Jones Substance). *A material modelled with the Lennard-Jones potential, whose interaction is set by only two parameters, σ and ϵ . These rescale the size and depth of the same fixed curve, so they change how large and how strongly bound the atoms are, but never the nature of the bonding. This restricts LJ to systems with genuinely simple, isotropic¹⁴ bonding, such as the noble gases.*

Noble gas atoms really do behave this way: spheres that weakly attract and repel when squeezed, with no preferred direction. Silicon does not. Its bonds demand specific angles, and LJ has no parameter for angles, so no choice of σ and ϵ can represent it.

2.3.2 Other Interatomic Potentials

Several families of potentials exist beyond LJ, each suited to different bonding environments:

¹⁴The same in all directions, with no preferred bonding angles.

- **Stillinger-Weber (SW)** [17]. Includes explicit three-body terms, well-suited for covalently bonded materials like silicon.
- **Bond order potentials.** Make bond strength depend on the number and type of neighbours. Examples include the Tersoff [18] and Brenner potentials, widely used for carbon-based materials.
- **Non-parametric / machine learning potentials.** No fixed functional form. The potential is instead learned from data. Examples include Moment Tensor Potentials and neural network potentials. These can capture complex interactions and are increasingly used for materials with non-trivial bonding¹⁵ such as a-Si.

2.3.3 Density Functional Theory

Definition 26 (Density Functional Theory). *A quantum-mechanical method for computing the ground-state energy and electronic structure of a system of interacting electrons, using the electron density rather than the many-body wavefunction.*

The many-electron Schrödinger equation is unsolvable for all but the smallest systems, since its wavefunction depends on the positions of every electron at once and its complexity explodes as electrons are added. The density $n(\mathbf{r})$ avoids this: it only describes how electrons are distributed in space, so it is a function of 3 coordinates rather than $3N$ for N electrons.

Hohenberg and Kohn [14] showed this is valid: the ground-state energy is fully determined by the density alone, and the true density is the one that minimizes it. The full wavefunction is never needed, and finding the density becomes an energy-minimization problem. In practice it is obtained through the Kohn-Sham scheme [15], which replaces many interacting electrons with non-interacting ones moving in an effective field chosen to reproduce the same density. The swap is not free: all the physics of how electrons truly interact gets pushed into one leftover term, the exchange-correlation functional.

Definition 27 (Exchange-Correlation Functional). *The exchange-correlation functional $E_{xc}[n]$ ¹⁶ collects all the many-body electron interactions (exchange and correlation) not captured by the simpler kinetic and electrostatic terms.*

Since E_{xc} has to be approximated, the accuracy of a DFT calculation depends on how good that approximation is. Two common ones are the local density approximation (LDA), which estimates E_{xc} from the electron density at each point, and the generalized gradient approximation (GGA) [16], which also looks at how quickly the density is changing from point to point.

¹⁵Non-trivial bonding refers to bonding that is not simply pairwise, but involves more complex interactions.

¹⁶This is the only part of DFT that is not known exactly and is responsible for most of the error in practical calculations.

The problem with DFT is its cost. The amount of computing time grows roughly as the cube of the number of electrons, so doubling the system size makes it about eight times more expensive. This quickly becomes impractical for the long simulations and large atom counts needed to model a-Si. The workaround is to run DFT only on a limited set of atomic arrangements to get accurate reference energies and forces, then train a potential like the MTP to reproduce those results at a tiny fraction of the cost.

2.3.4 Machine Learning Interatomic Potentials

Definition 28 (Machine Learning Interatomic Potentials). *Machine learning interatomic potentials (MLIPs) are computational models that use machine learning techniques to predict the potential energy and forces of atomic systems based on training data from high-fidelity calculations.*

2.3.5 Moment Tensor Potential

Definition 29 (Moment Tensor Potential). *Moment Tensor Potentials (MTPs) [8] is a specific type of MLIP that approximate DFT. Their accuracy can be dialed up in a controlled way, the model is built from a set of mathematical building blocks, and adding more of them makes it more accurate (though slower).*

Why MTP? Empirical potentials are fast but locked into a fixed formula, so their accuracy can only go so far. General machine-learning potentials are flexible but computationally expensive. MTP sits in between:

- **Systematically improvable:** its accuracy can be increased in a controlled way by adding more building-block functions.
- **Physically consistent:** it gives the same energy no matter how the atoms are labelled, rotated, or mirrored, as real physics should.
- **Linear scaling:** the cost grows only in proportion to the number of atoms, because each atom only interacts with those within a fixed local range.
- **Multi-element:** it can handle more than one atom type, with the interactions adjusted per chemical species.

Kokkos-accelerated Moment Tensor Potential Until recently the MTP had no GPU implementation, which limited how fast it could run on modern HPC¹⁷ systems. These machines are mixed, pairing normal CPUs with GPUs from NVIDIA, AMD, or Intel, and each type of hardware normally needs different programming methods to use efficiently. Meng et al. [5] solve this by extending the MTP inside the LAMMPS Kokkos package.

¹⁷High-Performance Computing systems are large setups of many connected machines, combining CPUs and GPUs to perform very large calculations.

Definition 30 (Kokkos). *A tool that lets one version of the code run on CPUs and on different GPUs without being rewritten for each device.*

The package provides three implementations. The first is an optimized CPU version (`mtp`), reaching up to a $2\times$ speedup over the earlier MLIP-3 code. The other two are GPU variants: a thread-parallel version (`mtp/kk`) for large simulations, typically 50,000 atoms or more per GPU, and a block-parallel version (`mtp/small/kk`) for smaller ones, typically 2000 atoms or more per GPU. To stay within GPU memory, both GPU variants use a `chunksize` parameter that splits atom processing into chunks so working memory can be reused. Cost is governed by the MTP level, which scales the number of basis functions exponentially.

The authors benchmarked on Narval, whose CPU nodes carry two AMD EPYC 7532 processors (32 cores each) and whose GPU nodes carry four NVIDIA A100 SXM4 (40 GB) cards. The GPU variants scale efficiently to million-atom systems. They stress that performance depends heavily on hardware, material, and simulation conditions, and recommend short single-GPU trial runs of both GPU variants, since the crossover at which the thread-parallel version overtakes the block-parallel one shifts with atom count, MTP level, and hardware.

2.4 Sampling Beyond MD

2.4.1 Activation Relaxation Technique Nouveau

Definition 31 (Activation Relaxation Technique Nouveau). *Activation Relaxation Technique nouveau (ARTn) [13] is a saddle-point search algorithm used to map the potential energy surface (PES)¹⁸ and identify transitions between local minima.*

Unlike straight MD, which has to wait for thermal fluctuations to drive a barrier crossing, ARTn explicitly climbs out of a minimum, finds a nearby saddle, and relaxes into a new minimum. It does this in three phases.

Activation Starting at a local energy minimum, the system is pushed in a random direction. After each push the lowest eigenvalue of the Hessian is checked (using the Lanczos algorithm, which only needs force differences and never builds the full Hessian). As long as this eigenvalue stays positive, the system is still inside the basin of the minimum, so it keeps pushing. Once the eigenvalue flips negative, a direction of negative curvature has been found, pointing toward a nearby saddle.

Relaxation The system is then moved along the negative-curvature direction while the energy is simultaneously minimized in all perpendicular directions. This converges on a first-order saddle point.

¹⁸Picture a landscape where height is how much energy a given atom arrangement has. Low valleys are stable arrangements, hilltops are unstable ones. To get from one stable arrangement to another, the atoms must pass through a higher-energy state in between, which is the hill.

Falling The system is nudged just past the saddle along the unstable direction and relaxed normally, settling into a new local minimum on the other side of the barrier.

Definition 32 (Activated Event). *A single transition in which the system leaves one local energy minimum, crosses a saddle point, and settles into another. “Activated” refers to the energy input needed to get over the barrier.*

Definition 33 (Metropolis Acceptance Probability). *A rule for accepting or rejecting a proposed move: always accept if it lowers the energy, otherwise accept with a probability that falls off exponentially with the energy increase,*

$$P_{acc} = \min \left[1, \exp \left(-\frac{\Delta E}{kT} \right) \right], \quad (2)$$

where ΔE is the energy difference between the old and new configurations, k is the Boltzmann constant, and T is a fictitious temperature.

Each ARTn cycle proposes one such event, accepted or rejected by Eq. 2 with T set to 0.25 eV in Zongo et al. [1]. A larger T accepts more uphill moves, which lets the search escape shallow traps instead of sticking in the first minimum it finds. This is what is meant by ARTn “capturing rare events”.

2.4.2 ARTn-MTP Models of Amorphous Silicon

Zongo et al. (2025) [1] is the central benchmark study for this project. It combines ARTn with the unified Si/O/SiO₂ MTP of Zongo et al. (2024) [2] to build seven a-Si models of 216 to 4096 atoms, and compares them against melt-quench MD with the same potential, experimental data, and the 100,000-atom 100k-GAP18 reference model [12]. The MTP training set contained no a-Si configurations, so these runs also test how well the potential transfers to the amorphous phase.

Most models were built by randomly filling a box at fixed density (2.20 or 2.28 g/cm³) and relaxing it with ARTn, which here acts as an energy minimizer that escapes local traps by climbing through saddle points. The starting configurations were very defective, around 70% wrong coordination in several cases.

Quantity	ARTn-MTP	Reference
Coordination defects	< 2% (3% for 1000-R, 4096-R-MD)	—
Angular distribution function	peaks at 109.5°, spread 10°	matches experiment
Ring statistics	peak at 5, 6, 7	no 3-rings, few 4-rings
Bulk modulus	~77 GPa	98.3 GPa (c-Si)
Excess energy	0.167 eV/atom	0.191 (MD), 0.135–0.205 (exp.)

Table 2: Zongo et al. [1] benchmark values for their ARTn-MTP a-Si models. The 216-R2 model reached 100% fourfold coordination, rarely seen outside a few Keating-potential WWW models.

Structure was checked with OVITO [20] for coordination (2.85 Å bond cut-off), R.I.N.G.S [21] for ring statistics under the King criterion, and ISAACS for the structure factor.

Definition 34 (Angular Distribution Function). *The distribution of bond angles θ across the structure, written $P(\theta)$. In a-Si it peaks at 109.5° with a spread near 10° .*

Definition 35 (Keating Potential). *An early empirical potential for tetrahedral semiconductors, penalizing only deviations in bond length and bond angle from their ideal values.*

Four models showed zero crystallinity, all four generated from a random start with no MD prerun. Three trends follow:

1. Crystallinity depends little on system size.
2. A random start gives less of it than a melt-quench start.
3. Fewer coordination defects tend to go with more crystallinity.

Definition 36 (Core-year). *One CPU core running for one year. The standard unit for reporting the cost of an HPC calculation, since it does not depend on how many cores ran in parallel.*

The cost is modest: 0.0615 core-years for the 216-atom model and 0.34 for the 1000-atom one.

The conclusion is that ARTn-MTP gives low crystallinity and low defects together, while MD melt-quench traps crystalline grains, making ARTn-MTP the better route to near-ideal CRN structures.

2.4.3 Morphological Autoregressive Protocol

Definition 37 (Morphological Autoregressive Protocol). *The Morphological Autoregressive Protocol (MAP) is a generative machine-learning method that builds large models of disordered matter at a cost proportional to system size, rather than exponential in it.*

Definition 38 (Autoregressive). *An autoregressive model is one where the prediction for the next value depends only on the previous values.*

Definition 39 (Correlation Length). *The correlation length L_c is the distance over which the influence of one atom on another fades out. It is read off from the radial distribution function.*

The structure is chopped onto a grid and built one point at a time, like generating a sequence word by word. For each grid point the model learns the probability that it holds a given atom type, based only on the points already placed within L_c :

$$P(X_i = \theta \mid \{X_{j < i}\}) = \mathcal{N}[c_{b,\theta} + F_\theta(\{X_j \in L_c\})], \quad (3)$$

where X_i is the grid point being filled, θ is the atom type, and $\mathcal{N}$ is a softmax turning raw scores into probabilities that sum to one. Feeding each sampled point back in as context extends a small training sample to any size, at linear cost.

The grid cells are cubes, or voxels. A 0.5 Å voxel, a quarter of the Si-Si bond length, gives 88^3 cells for a 44 Å box.

Definition 40 (Hyperparameter). *A setting chosen before training rather than learned from the data, such as the voxel size, the number of convolutional layers, or the number of training epochs. Values are picked by trial and comparison rather than by fitting.*

Definition 41 (PixelCNN). *A PixelCNN is a convolutional neural network with masked filters, allowing each pixel to see only previously generated pixels.*

Definition 42 (Markov Chain). *A Markov chain is a sequence of states where each state depends only on the previous one.*

F_θ is a function that maps the local environment to the probability of an atom of type θ , and is learned with a PixelCNN, extending an earlier 2D single-element version (Kilgour et al., 2020 [6]) to 3D with multiple atom types. Repeated generation forms a Markov chain that settles onto the training distribution, which is what makes the output physically meaningful rather than random.

The method was tested on liquid water. The generated samples reproduce the reference radial distribution functions, bond angles and bond lengths, and are stable enough to seed MD runs, so placing atoms on a grid does not corrupt the physics.

MAP as an Anomaly Detector Madanchi and Simine (2026) [4] use the same trained model in reverse: instead of generating structures, they ask whether the uneven dynamics of a supercooled liquid¹⁹ can be predicted from a single static snapshot, with no information about motion. MAP already knows what a normal local arrangement looks like, so it can flag unusual ones without being told during training which regions moved.

Definition 43 (Dynamic Heterogeneity). *In a supercooled liquid, some regions rearrange quickly while others stay frozen, even though the material looks uniform. The motion is uneven in space rather than spread evenly.*

Definition 44 (80:20 Kob-Andersen Mixture). *A standard model glass-former: a binary mixture of two particle types with different interaction parameters, in an 80:20 ratio.*

Definition 45 (h-excitation). *A dynamic excitation identified directly from an MD trajectory: a particle whose positions before and after a time window t_a differ by more than a displacement threshold a . It marks a discontinuous jump to a new location, and serves as the ground truth against which Ω is tested.*

¹⁹A liquid cooled below its freezing point without crystallizing, on its way to becoming a glass.

The test system is the 80:20 Kob-Andersen mixture. They define a per-particle score Ω , the average of six probabilities, one for each of six fragmented views C_i of the particle’s surroundings:

$$\Omega_j = \frac{1}{6} \sum_{i=1}^6 p(\mathbf{r} \mid C_i). \quad (4)$$

Low Ω means the particle sits in an unlikely arrangement, a sign it is about to rearrange or is rearranging now. High Ω means the opposite.

Comparing Ω against the h-excitations²⁰, about 88% show up as a dip in Ω and the rest as a bump. Pushing the threshold close to the baseline catches nearly all of them but also flags particles that are not visibly moving; more restrictive thresholds reduce the agreement to about 84% while cutting those false positives. The flagged non-movers turn out to have an unusual number of already-moving neighbours, so Ω is picking up activity around the particle as well as in it. This gives an interpretable way to spot structural warning signs of motion with no labelled examples, useful for catching unusual regions in melt-quench simulations.

2.5 Structural and Dynamical Characterization

An MD simulation using MLIPs produces nothing but a list of atomic positions at each timestep. Deciding whether that list represents a crystal, a liquid, or an amorphous solid requires reducing it to a small number of quantities that distinguish the three phases. Four are used in this project: the radial distribution function, the coordination number, the mean square displacement, and the tetrahedral order parameter. The first two describe structure, the third describes dynamics, and the fourth measures local geometry directly.

2.5.1 Radial Distribution Function

Definition 46 (Radial Distribution Function). *The radial distribution function (RDF), written $g(r)$ ²¹, gives the probability of finding an atom at a distance r from a reference atom.*

Remark 3. *Basically, pick an atom, look outward, count everything you find at each distance and it is averaged over every atom and every frame.*

The value of $g(r)$ tells you how likely you are to find an atom at separation r compared to a uniform gas of the same density.

- $g(r) > 1$, atoms are more likely than chance to sit at that distance, producing peaks.

²⁰The “h” refers to the Heaviside step function used to model the discontinuous jump in position.

²¹Incidentally, $g(r)$ doubles as a bridge to two other things: the distance at which its oscillations die out is the correlation length L_c that MAP relies on, and its Fourier transform is the static structure factor $S(q)$ that X-ray and neutron diffraction experiments measure directly.

- $g(r) = 1$, an atom is just as likely to be there as it would be by chance.
- $g(r) < 1$, that separation is avoided.
- $g(r) = 0$, no atom is ever found there. This is the case at small r , where the atoms would be so close that their electron clouds overlap and repel.

Formally, for a system of N atoms at number density $\rho = N/V$,

$$g(r) = \frac{1}{4\pi r^2 \rho N} \left\langle \sum_{i=1}^N \sum_{j \neq i} \delta(r - r_{ij}) \right\rangle, \quad (5)$$

where r_{ij} is the separation between atoms i and j , the angle brackets denote a time average over the trajectory, and the factor $4\pi r^2$ is the surface area of the spherical shell at radius r . Dividing by this shell area is what converts a raw count of neighbours into a density, since a shell far from the reference atom is larger and would otherwise contain more atoms.

The shape of $g(r)$ is the primary structural fingerprint of a phase:

Phase	1st peak	2nd peak	Higher shells
c-Si	sharp, 2.35 Å	sharp, 3.84 Å	all present and sharp, never decay
l-Si	broad, 2.45 Å	weak smear, 3.8 Å	none, flat past ~5 Å
a-Si	sharp, 2.35 Å	present, 3.84 Å	3rd weak or absent, flat past 7–8 Å

Table 3: Radial distribution function signatures of the three phases.

The second peak comes from the tetrahedral angle of 109.47° being preserved locally, so a sharp second peak is what separates an amorphous network from a liquid. Losing the higher shells is what separates it from a crystal, where the peaks persist undamped: that persistence is what long-range order means.

Two different cutoffs. Two distances get called cutoffs here and they do unrelated jobs.

- The first is the potential’s cutoff, 5.7 Å for the MTPs used. During the run LAMMPS computes no force between atoms further apart than this.
- The second is the range over which $g(r)$ is measured. This happens after the run, on the saved positions, and only measures pair distances, so nothing prevents measuring a pair 8 Å apart; they simply never interacted directly. Its limit comes from the periodic boundaries instead: past half the box width you start measuring an atom against copies of itself. The 216-atom box is 16.4 Å wide, giving 8.2 Å.

This raises an obvious question. If forces stop at 5.7 Å, why does $g(r)$ show structure past that? Because constraint propagates down the chain: A pushes B, B pushes C, C pushes D. By the time you reach D you are well beyond 5.7 Å from A, but D’s position is still constrained, just indirectly. That is how a crystal holds long-range order with a short-range potential.

2.5.2 Coordination Number

Definition 47 (Coordination Number). *The coordination number is the average number of nearest neighbours per atom.*

The running coordination number follows from $g(r)$ by integration over the first neighbour shell,

$$n(r) = 4\pi\rho \int_0^r g(r') r'^2 dr', \quad (6)$$

Definition 48 (Coordination Number Proper). *The coordination number proper is $n(r_{cut})$ evaluated at the first minimum of $g(r)$, which is the natural boundary between the first and second neighbour shells²².*

The coordination number separates the three phases cleanly:

- **c-Si**: exactly 4, by construction.
- **a-Si**: close to 4. The deviation is the coordination-defect concentration of Section 5: threefold atoms are dangling bonds, fivefold atoms are floating bonds.
- **l-Si**: roughly 6. Melting destroys the directional covalent bonding and the liquid adopts a denser, more metallic packing.

That last change is why liquid silicon is denser than the crystal (2.57 against 2.33 g/cm³), an anomaly shared with water. It also gives an independent thermodynamic check on melting: under NPT the volume should contract, not expand, when the crystal melts.

2.5.3 Mean Square Displacement

Definition 49 (Mean Square Displacement). *The mean square displacement (MSD) measures how far atoms have travelled from their initial positions, averaged over all atoms.*

$$MSD(t) = \frac{1}{N} \sum_{i=1}^N \left\langle |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \right\rangle. \quad (7)$$

Unlike $g(r)$ and the coordination number, the MSD is dynamical and requires a trajectory rather than a single configuration. In a solid, atoms vibrate about fixed sites, so the MSD rises briefly and plateaus near 0.1 Å², set by the thermal vibration amplitude. The plateau grows with temperature but stays bounded. In a liquid, atoms diffuse freely, so after a short ballistic transient the MSD grows linearly and without bound, with the slope giving the self-diffusion coefficient through the Einstein relation

$$D = \lim_{t \rightarrow \infty} \frac{MSD(t)}{6t}, \quad (8)$$

²²For silicon this cutoff is conventionally taken as 2.85 Å, the value used by Zongo et al. in their OVITO coordination analysis.

where the factor 6 is $2d$ for $d = 3$. A linear, unbounded MSD is therefore the most direct confirmation that a crystal has melted.

2.5.4 Tetrahedral Order Parameter

Definition 50 (Tetrahedral Order Parameter). *The tetrahedral order parameter q measures how closely an atom’s four nearest neighbours approach ideal tetrahedral geometry:*

$$q = 1 - \frac{3}{8} \sum_{j=1}^3 \sum_{k=j+1}^4 \left(\cos \theta_{jk} + \frac{1}{3} \right)^2, \quad (9)$$

where θ_{jk} is the angle at the central atom between neighbours j and k , summed over all six distinct pairs.

The form of Eq. 9 is fixed by its two limits. In a perfect tetrahedron every bond pair subtends 109.47° , so $\cos \theta = -1/3$, every bracket vanishes, and $q = 1$. For randomly oriented neighbours the sum averages to $8/3$, so the $3/8$ prefactor makes $\langle q \rangle = 0$.²³ The parameter therefore runs from 1 for ideal tetrahedral order to 0 for a structureless arrangement, with no reference table required. This normalization matches Zhao et al. [22], who applied q to *ab initio* molecular dynamics of liquid silicon.

Crystalline silicon sits at $q = 1$ by construction. Amorphous silicon falls slightly below, since the network keeps fourfold coordination but broadens the bond-angle distribution. Liquid silicon falls further as coordination rises toward six and tetrahedral geometry is lost.

2.6 Computational Environment

2.6.1 Narval

Definition 51 (Narval). *Narval is a general purpose cluster designed for a variety of workloads. Built by Dell Canada and CDW Canada it is located at the École de technologie supérieure in Montreal. The cluster is named in honour of the narwhal, a species of whale which has occasionally been observed in the Gulf of St. Lawrence.*

Compiling on a Compute Node. Compilation²⁴ is CPU-intensive and belongs on a compute node rather than the shared login node, which is reserved for lightweight tasks. A node is requested from the SLURM scheduler with an interactive `salloc` allocation; once granted, the prompt changes from `narval1` to a compute node such as `ng10403`, and the software environment is loaded before the build script is run. None of the required compilers or libraries are available until explicitly requested with `module load`. The exact commands are given in Appendix E.4.

²³Random orientation gives $\langle \cos \theta \rangle = 0$ and $\langle \cos^2 \theta \rangle = 1/3$, so each bracket averages to $1/3 + 1/9 = 4/9$ and the six pairs sum to $8/3$.

²⁴Compilation is the process of converting source code into an executable program.

Filesystems and Their Quotas. Narval exposes several filesystems with different purposes and quotas, and the distinction matters for long production runs. `/home` is small and backed up; `/scratch` is large, fast, per-user and *purged* on a fixed retention schedule; `/project` is the group allocation, shared between all members and quota-limited on both total size and file count; `/nearline` is tape-backed archival storage. Current usage against quota is reported by `diskusage_report`. All production runs in this project were executed from `/scratch`, with finished configurations copied to `/project` for retention.

2.6.2 LAMMPS

Definition 52 (LAMMPS). *Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS), developed and distributed by Sandia National Laboratories, is an open-source code for classical particle-based materials simulation. It is the molecular dynamics code used in this project, run on a remote computing cluster accessed via SSH (PuTTY).*

Simulation Workflow. A LAMMPS run in this project involves three files:

- The input script (with `.in`, `.melt`, and `.imp` as file extensions) is the recipe written in LAMMPS’ command syntax.
- The shell script (with `.sh` as its file extension) is a short bash script that launches LAMMPS and feeds it the input script, typically through a line of the form `‘imp_serial -in test.in’`²⁵.
- The log file (`log.lammps`) is generated automatically by LAMMPS during the run and records the thermodynamic output (temperature, energy, and pressure at each step), along with timing information and any error messages.

Combining LAMMPS with the MTP. LAMMPS handles the mechanics of a simulation (tracking atomic positions, building neighbour lists, and integrating Newton’s equations of motion) but it is agnostic about where the forces come from. The interatomic potential is supplied as a separate, interchangeable module selected through the `pair_style` command. Combining LAMMPS with the MTP therefore means installing an MTP pair style into LAMMPS. Once built, the trained potential is used by pointing the input script at the model file, e.g. `pair_style mtp model.almtp`: at every timestep, LAMMPS passes each atom’s local neighbourhood to the MTP, which returns the energies and forces that LAMMPS then uses to advance the system. The division of labour is clean: the MTP decides how atoms interact, and LAMMPS decides what happens next.

²⁵The word ‘serial’ indicates the run used a single processor core with no parallelism, which is appropriate for a small test case.

2.6.3 WinSCP

Definition 53 (WinSCP). *WinSCP is a free and open-source software (application) for Microsoft Windows. Its main function is secure file transfer between a local and a remote computer.*

When logged in to WinSCP using the login node ‘narval.alliancecan.ca’ and the username and password created through the CCDB website, enter the path ‘/scratch/your_username/'. Then you can drag and drop files between your local computer and the Narval cluster. Do not click on the scratch folder in the home menu, since it will load all folders of every Narval user.

2.6.4 PuTTY

Definition 54 (PuTTY). *PuTTY is a free and open-source terminal emulator and serial console. It supports various network protocols, including SSH, Telnet, rlogin, and raw socket connections. In this project, PuTTY is used to establish a secure SSH connection to the Narval cluster for remote command-line access.*

PuTTY uses the same login credentials as WinSCP, but instead of dragging and dropping files, it lets you run commands directly on the cluster. It behaves like a standard Linux terminal: you navigate the filesystem, edit files, and launch simulations by typing commands. The typical workflow is to move into your scratch folder, then submit or run a LAMMPS job from there.

Interactive Jobs. A regular batch job (`sbatch`) submits a fixed script to the queue and runs unattended; the user only sees the output afterwards. An interactive job instead gives direct command-line access to the compute node for the duration of the allocation, so commands can be typed, tested, and corrected in real time. This makes it the right tool for tasks that require trial and error, such as compiling software, debugging a crashing binary, or checking that modules load correctly, where waiting through the batch queue after every small change would be far too slow. Batch jobs remain preferable for long production runs, since they do not require the user to stay connected. In this project, interactive jobs were used to compile the custom LAMMPS-MTP build on a compute node (Section 2.6.1).

Definition 55 (Interactive Job). *An interactive job is a compute-node session requested through the scheduler with `salloc`, which drops the user into a live shell on the allocated node instead of running a pre-written script.*

2.6.5 Visual Molecular Dynamics

Definition 56 (Visual Molecular Dynamics). *Visual Molecular Dynamics (VMD) is a molecular visualization program for displaying, animating, and analyzing large biomolecular systems using 3D graphics and built-in scripting.*

In this project, VMD is used to visualize the atomic configurations and trajectories generated by LAMMPS. A practical subtlety is that LAMMPS produces two distinct kinds of output, and VMD loads each differently. A data file (such as `config.end_sim`, written by `write_data`) stores a single configuration, whereas a trajectory file (such as `dump.lammpstrj`, written by the `dump` command) stores many frames. The graphical loader only exposes the trajectory reader, so data files must instead be loaded through the Tk Console using the TopoTools plugin.

Once a structure is loaded, the drawing method is set to VDW so atoms render as spheres, and `pbx box` draws the periodic simulation cell. For a trajectory, the playback controls in the main VMD window then animate the configuration through all stored frames.

2.6.6 OVITO and ISAACS

Definition 57 (OVITO). *The Open Visualization Tool [20], used here for coordination analysis and for the “identify diamond structure” crystallinity test.*

Definition 58 (ISAACS). *A structure-analysis package for amorphous and crystalline systems, used by Zongo et al. to compute the structure factor.*

3 Methodology

3.1 Toolchain Validation: Lennard-Jones Melt

Before simulating a-Si, a standard Lennard-Jones (LJ) melt was run on Narval to confirm that the full toolchain (job submission, LAMMPS execution, and output writing) functioned correctly. This is a generic LJ fluid and not a-Si; its only purpose is validation. The input script can be found in `code/lj_toolchain_tests/test.in`.

3.1.1 Output and Validation

The total energy²⁶ per atom remained essentially constant at approximately -4.62 throughout the run. Because the NVE ensemble conserves energy, this constancy confirms that the integration was stable and the simulation parameters were sound. The temperature²⁷ relaxed from its initial value of 1.44 to a steady value near 0.70 as the initial lattice melted and equilibrated. The run completed 10,000 steps on 13,500 atoms in roughly 95 seconds of wall time, with no dangerous neighbour-list builds reported, confirming a working toolchain.

3.1.2 VMD visualisation

The next step was to visualize the completed LJ melt. The trajectory file `dump.lammpstrj` produced by the run was transferred to the local machine with WinSCP and loaded into VMD, giving a 3D animation of 101 frames (the initial configuration plus one frame per 100 timesteps). The animation shows the initial fcc lattice melting into a disordered fluid, shown in Fig. 3.

²⁶Here energy is negative to signify it would cost energy to pull this system apart.

²⁷In reduced LJ units, temperature is dimensionless (measured in units of ϵ/k_B).

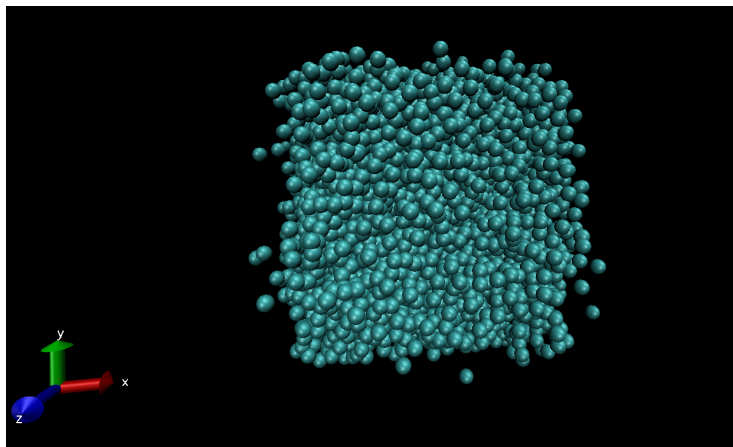


Figure 3: Visualization of the Lennard-Jones melt simulation in VMD. The atoms are displayed as spheres, and the system has equilibrated into a disordered fluid state.

3.2 Building LAMMPS with the Moment Tensor Potential

The Lennard-Jones test used the standard LAMMPS installation already provided on Narval (loaded as a module). Simulating silicon with an MTP instead requires a custom build, because the mtp pair styles are not part of standard LAMMPS. The code was provided by Alliance technical support as an archive, ‘LMP_MTP.tar.gz’, placed in the home directory. It bundles the LAMMPS source, the MTP source files, and two build scripts: one for a CPU-only build (‘build-lmp-mtp-cpu.sh’) and one for the GPU (Kokkos) build (‘build-lmp-mtp-gpu.sh’).

3.2.1 Debugging the CPU Build

Symptom	Cause	Fix
Illegal instruction (core dumped)	Binary compiled for a CPU architecture incompatible with Narval's nodes	Add <code>-DCMAKE_CXX_FLAGS="-march=core-avx2"</code>
ML-MTP absent from <code>lmp -h</code>	Inverted conditional (<code>if [! -d ...]</code>) skipped the source copy	Correct the conditional logic by removing the <code>!</code>
Rebuild produced no new binary	Stale nested CMake caches with another user's paths baked in; <code>--fresh</code> resets only the top level	Delete <code>_build</code> entirely before rebuilding

Table 4: CPU build faults in the supplied LAMMPS-MTP archive and their fixes.

After a clean rebuild, `lmp -h` listed ML-MTP with `mtp` and `mtp/extrapolation` available as pair styles. Validation was a 10,000-step, 13,500-atom Lennard-Jones melt submitted through SLURM, completing in 4 min 16 s of wall time with no dangerous neighbour-list builds.

3.2.2 Debugging the GPU Build

Symptom	Cause	Fix
<code>mtp/kk</code> styles absent	Script copies only ML-MTP; the KOKKOS folder was never copied	Copy it into <code>src/KOKKOS/</code>
Binary would not run on GPU	Kokkos CUDA backend not enabled	<code>kokkos-cuda</code> preset + <code>-DKokkos_ARCH_AMPERE80=ON</code> (A100)
Compilation errors under NVIDIA compiler	<code>-march=core-avx2</code> inherited from the CPU script	Remove the flag
GPU package build failure	OpenCL-based <code>-DPKG_GPU=ON</code> , redundant alongside Kokkos	Remove the option

Table 5: GPU (Kokkos) build faults and their fixes.

The GPU build then compiled with `mtp/kk` and `mtp/extrapolation/kk` present, and ran the same Lennard-Jones test on an A100 node in 15.6 s against

4 min 16 s on CPU, a speedup of roughly $16\times$.

3.3 Simulation Protocol for Amorphous Silicon

Ata provided five silicon MTPs for this project, split into two groups. Two are melt-stable (levels 22 and 26), fitted explicitly to remain robust at the high temperatures of a melt. The other three are pruned potentials (**pruned_1**, **pruned_2**, **pruned_3**), ordered by increasing computational cost and accuracy, but potentially unstable in a melt. The a-Si is produced by a melt-quench protocol:

1. **Melt.** Heat a c-Si lattice to about 2500 K with a melt-stable potential, and save a snapshot.
2. **Hold.** Hold the melt at temperature for 1 ns to erase memory of the crystal.
3. **Quench.** Cool from the melt temperature to 300 K, fast enough that the liquid structure freezes in as a-Si rather than recrystallizing.
4. **Equilibrate.** Hold at 300 K for 150 ps to relax the structure and remove any residual stress.

All production runs use the CPU build. The 216-atom cell is well below the ~ 2000 atoms per GPU at which even the block-parallel GPU variant becomes worthwhile, so the GPU build validated in Section 3.2 was not used beyond that validation.

3.3.1 Atom Type Mapping

The potential file declares `species_count = 2` without stating which type is which element. This was settled by two NVT runs with potential `22.almtp` on 216 atoms at 300 K in a fixed box of $4433 \text{ \AA}^3$. Assigning the atoms to type 1 gave a stable run at a density of 2.27 g/cm^3 , close to the measured 2.33 g/cm^3 for c-Si. Assigning them to type 2 gave a starting pressure of $-2.24 \times 10^6 \text{ bar}$ and the box collapsed within a few steps into a neighbour-list overflow. Type 1 is therefore silicon and type 2 oxygen, consistent with a potential trained on the Si/O/SiO₂ system.

3.3.2 First Silicon Runs and Choice of Ensemble

Both ensembles (NVT and NPT) were tested on a 216-atom diamond lattice ($3 \times 3 \times 3$ unit cells, $a = 5.431 \text{ \AA}$, periodic, `units metal`, 1 fs timestep). The intended workflow was NPT equilibration to find the box size the potential prefers, followed by NVT production at that fixed volume.

The protocol was subsequently fixed to NPT only, following Zongo et al. [1], so the NVT stage was dropped. The runs below are retained here as a benchmark of the potentials and of parallel scaling.

The NPT test was run at 300 K and 0 bar and expanded the box from 4325 to 4433 Å³, corresponding to a lattice parameter of $a = 5.476$ Å. This is 0.8% above the measured value of 5.431 Å, an overestimate consistent with the PBE²⁸ [16] reference data used to train the potential.

The NVT test was then performed with the lattice rebuilt at $a = 5.476$ Å, the volume stayed locked, the pressure settled to order 10²–10³ bar rather than the several-thousand-bar swings seen under NPT, and the potential energy showed no drift.

Potential	Ensemble	Cores	Pair time (s)	Wall time
22.almtp	NVT	1	1374.7	22:55
pruned.1	NVT	8	12.6	00:14
22.almtp	NPT	1	1222.6	20:24

Table 6: Cost of 5000 steps on 216 atoms. Force evaluation dominates throughout ($\geq 92\%$ of loop time) and no neighbour-list rebuilds occurred in any run.

The two potentials prefer different volumes. An unexpected result came out of the NVT tests. With `22.almtp`, the pressure at a box volume of 4433 Å³ fluctuated around zero, meaning the atoms sit at a spacing that potential is content with. With `pruned.1.almtp` in the identical box, the pressure instead held steady at about −12,400 bar. A negative pressure means the atoms are pulling inward: this potential would prefer a smaller box than the one it has been given.

3.3.3 Parallel Scaling

The single-core timings from the previous runs made the cooling stage impractical, so the next step was to find out how much of that cost could be removed.

The same 5000-step NVT run was repeated on 1, 4, and 8 cores of a single node. Running in parallel requires some changes to the submission script, such as launching LAMMPS through `srun` rather than calling it directly, so that SLURM starts one copy of the program per task and connects them through MPI.

²⁸Perdew-Burke-Ernzerhof, the generalized gradient approximation exchange-correlation functional used to generate the MTP training data.

Potential	Ensemble	Cores	Pair time (s)	Wall time	Speedup	APC ²⁹
22.almtp	NVT	1	1374.7	22:55	1.0×	216
22.almtp	NVT	4	305.9	05:07	4.5×	54
22.almtp	NVT	8	153.8	02:43	8.4×	27

Table 7: Wall time for a 5000-step NVT run of 216 silicon atoms using 22.almtp, on one node.

The scaling is close to linear up to 8 cores. The limit is nonetheless in sight: the time spent on communication between cores rose from 0.4% of the total at 4 cores to 4% at 8, and with only 27 atoms per core the workload is becoming too thin to divide further. Eight cores was therefore adopted as the default for systems of this size.

The core count is capped by the potential. Each core is given its own chunk of the box. To compute the forces on atoms near the edge of its chunk, a core also needs to know about atoms sitting just outside it, so neighbouring cores send copies across. How far outside it has to look is set by the potential’s cutoff, listed as `max_dist` in the .almtp files, which is 5.7 Å here. If a chunk is narrower than that, the core needs atoms from beyond its immediate neighbours, and LAMMPS cannot supply them. The box is 16.4 Å wide, so 64 cores would give 4 Å chunks and fail. Eight cores give 8.2 Å chunks, comfortably above the cutoff.

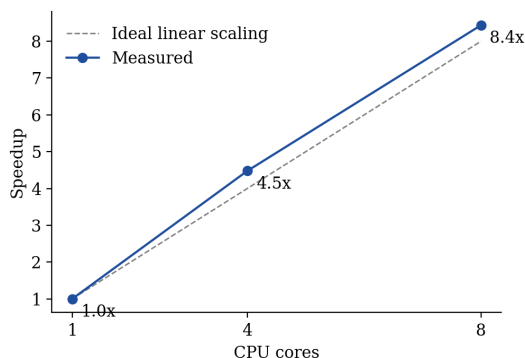


Figure 4: Parallel speedup of the 5000-step NVT run against the ideal linear case. Scaling is near-linear to 8 cores; the superlinear value at 4 cores likely reflects cache effects at small per-core atom counts.

3.3.4 Setting the Cooling Rate

A question had to be settled before the melt could be run: how to impose a cooling rate in LAMMPS.

LAMMPS has no command that sets a cooling rate directly. Instead, `fix npt` takes a starting and a target temperature, and ramps linearly between them over the course of the `run` that follows. The rate is therefore implied by the length of that run rather than specified:

$$\text{rate} = \frac{\Delta T}{N_{\text{steps}} \times \Delta t}, \quad (10)$$

where Δt is the timestep and ΔT the difference in temperature. Using Zongo’s cooling rate of 1 K/ps from 2500 K to 300 K, the temperature drop of 2200 K implies 2200 ps of simulated time, which at a 1 fs timestep is 2.2×10^6 steps.

An immediate consequence, which matters when a run has to be resumed, is that the ramp is defined relative to the current `run` command and not to any absolute timestep. Restarting a partially completed quench therefore requires the starting temperature and the remaining step count to be recomputed by hand from the point of interruption.

3.3.5 Projected Cost

Throughput on 8 cores follows directly from the 5000-step benchmarks of Table 6 and 7:

$$22.\text{almtp} : \frac{5000 \text{ steps}}{163 \text{ s}} = 30.7 \text{ steps/s}, \quad \text{pruned_1} : \frac{5000 \text{ steps}}{14 \text{ s}} = 357 \text{ steps/s},$$

where 163 s and 14 s are the measured wall times (2:43 and 0:14). The pruned potential is therefore $357/30.7 \approx 12$ times cheaper per step. At a 1 fs timestep, each stage of the protocol converts to a step count and a wall time as $t_{\text{wall}} = N_{\text{steps}}/r$:

Stage	Potential	Simulated time	Steps	Estimated wall time
Heating (300–2500 K)	22.almtp	30 ps	3.0×10^4	16 min
Hold at 2500 K	22.almtp	1 ns	1.0×10^6	9.1 h
Cooling (1 K/ps)	pruned_1	2.2 ns	2.2×10^6	1.7 h
Total		3.2 ns	3.2×10^6	11 h

Table 8: Projected cost of the melt-quench protocol for 216 atoms on 8 cores, using the melt-stable potential for heating and the hold and `pruned_1` for cooling. Cooling from 2500 K to 300 K at 1 K/ps fixes the cooling stage at 2200 ps.

The hold at 2500 K dominates despite involving less than half the steps of the cooling stage, purely because of the factor of twelve in cost per step. Whether the hold genuinely requires the melt-stable potential, or whether the switch to a pruned one can happen earlier, has been raised with the group. These figures also assume the cost per step stays as measured on a static 300 K crystal; a hot liquid rearranges continuously and will trigger neighbour-list rebuilds that the earlier runs did not (all benchmark runs recorded zero rebuilds), so the true cost will be somewhat higher.

3.4 Running MAP

MAP was installed to `~/MAP-main` with a virtual environment at `~/envMAP`, requiring `python/3.11` and `scipy-stack`.

Inspecting the bundled water dataset settled the data format. The files are `water.npy`, shape $(1000, 648, 3)$, and `water_identity.npy`, shape $(1000, 648)$ of species strings. MAP is not given a voxel grid; it voxelizes internally. The a-Si equivalent is therefore just $(500, 216, 3)$ coordinates paired with a $(500, 216)$ array of 'Si'.

The code writes checkpoints and reads data from hardcoded paths relative to the working directory, which would scatter output into the quota-limited home filesystem. Runs are executed from `/scratch` with `training_samples` symlinked in and the `ckpts/` prefix stripped from `save_ckpt`.

A validation run on unmodified water data was killed by the out-of-memory killer. This is host RAM rather than GPU memory. The fix was to request `--mem-per-gpu=100G`, matching the configuration used by the MAP authors, after which training ran to completion: 81 epochs at roughly 940 s each, with the test negative log-likelihood falling from 0.0102 to 0.0069 and flattening past epoch 58. Generation of a single sample then ran its full 80 steps in 1 h 17 min before failing on the last line, where `np.save` writes to the relative path `samples/run.0.npy` and the directory does not exist.

This is the same fault as the `ckpts/` prefix and points at a general property of the code: every output path is relative to the working directory and none is created if absent. The fix was to write the sample to an absolute path under `/scratch`. Since no checkpoint is written for a completed training stage, recovering the sample required repeating all 81 epochs, so a dry run at low `max_epochs` is worth doing after any change to the output paths.

Item	Current	Required for a-Si
Data filenames	water	a-Si equivalents
Augmentation slices	Index 40, assuming 80^3	~ 34 per side at 0.5 Å
<code>max_epochs</code>	80	500

Table 9: Changes required to train MAP on a-Si.

The voxel size remains undecided between 0.25 and 0.5 Å. The larger value follows the quarter-bond-length rule, but Fig. 9 shows only $\sim 1.3 \times 10^{-3}$ of atom-frames exceed 0.5 Å, so most atoms never leave their voxel and successive training frames are near-identical. The choice will be made on training loss and sample quality.

4 Results

Every run below uses the same 216-atom cell on 8 MPI tasks under NPT at zero pressure with a 1 fs timestep. The melt heats a diamond lattice from 300 to 2500 K over 30 ps, the hold keeps it there for 1 ns, and the quench cools it to 300 K at 1 K/ps followed by 150 ps of equilibration.

4.1 Melt

Both melt-stable potentials melted the lattice successfully. Melting is confirmed by the agreement of all five diagnostics of Section 2.5 rather than by any one alone, computed over the first 2 ps of the ramp (crystal) and the final 3 ps (liquid).

Quantity	22.almtp		26.almtp	
	crystal	liquid	crystal	liquid
$g(r)$ peaks (Å)	2.37, 3.87, 4.54	2.45	2.37, 3.87, 4.54	2.44
Coordination $\bar{n}$	4.000	5.968	4.000	5.964
Tetrahedral order q	0.9925	0.4977	0.9923	0.4905
standard deviation	0.0058	0.2046	0.0060	0.2080
MSD (Å ²)	< 1	111.4	< 1	107.5
Volume (Å ³)	4500	4310	4507	4326
Density (g/cm ³)	2.240	2.337	2.235	2.329

Table 10: Melting signatures for the 216-atom system heated to 2500 K. Crystal values are averaged over the first 2 ps of the ramp, liquid values over the final 3 ps. Coordination uses a 3.15 Å cutoff, the first minimum of the liquid $g(r)$. Density follows from $\rho = 216M_{\text{Si}}/(N_A V)$ with $M_{\text{Si}} = 28.0855$ g/mol.

Every row is a phase signature in its own right.

Radial distribution function. In the crystal, $g(r)$ shows sharp peaks at the diamond-lattice neighbour shells that stay just as sharp all the way out to the 8 Å box limit, with $g(r)$ falling to zero in the gaps between them: an atom at 3 Å from another simply does not occur³⁰.

In the liquid all of it is gone but the first shell. A single broad peak sits at 2.45 Å, and past about 3.5 Å the curve flattens onto $g(r) = 1$, meaning that beyond one bond length an atom is as likely to be found anywhere as anywhere else.

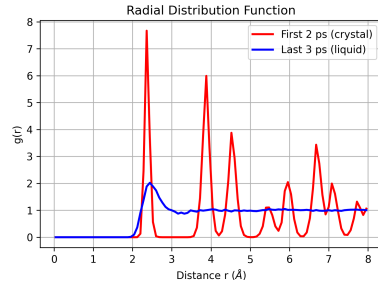
³⁰That is what long-range order looks like.

Coordination number. Coordination rises from exactly four to almost six. In the crystal each silicon is sp^3 bonded to four neighbours at fixed tetrahedral angles, and those directional bonds hold the atoms apart. Melting breaks them, and without the angular constraint the atoms pack together like a simple metal, which fits more neighbours into the same first shell.

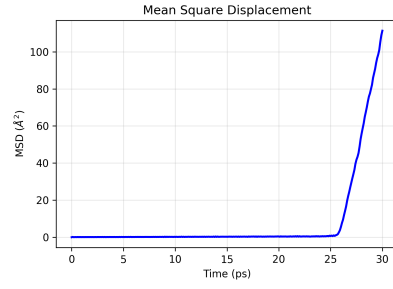
Mean square displacement. The MSD separates the two phases outright. For the first 25 ps it stays below $1 \text{ \AA}^2$, rising only as the atoms vibrate harder with temperature, since a vibrating atom stays near its lattice site no matter how hot it gets. At 25.6 ps it turns sharply and climbs in a straight line to $111.4 \text{ \AA}^2$ over the remaining 4.4 ps of the ramp ($107.5 \text{ \AA}^2$ from 25.7 ps for level 26). Unbounded linear growth means the atoms are diffusing, not vibrating.

Tetrahedral order parameter. The tetrahedral order parameter loses half its value and its distribution broadens by a factor of 35, the local-geometry counterpart of the loss of order in $g(r)$.

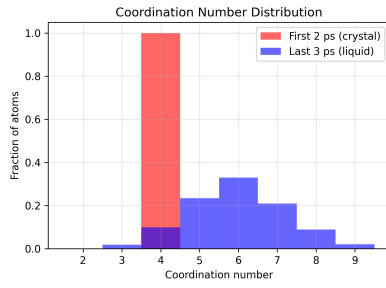
Density. The box contracts by 4.2% and 4.0% respectively, which is the correct sign for silicon: unlike most materials, liquid silicon is denser than the crystal, so melting must shrink the cell rather than expand it.



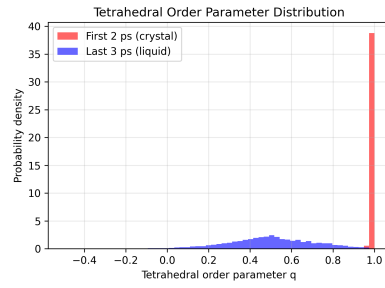
(a) Radial distribution function



(b) Mean square displacement



(c) Coordination number



(d) Tetrahedral order parameter

Figure 5: Melt with 22.almtf. Red is the first 2 ps of the ramp, blue the final 3 ps.

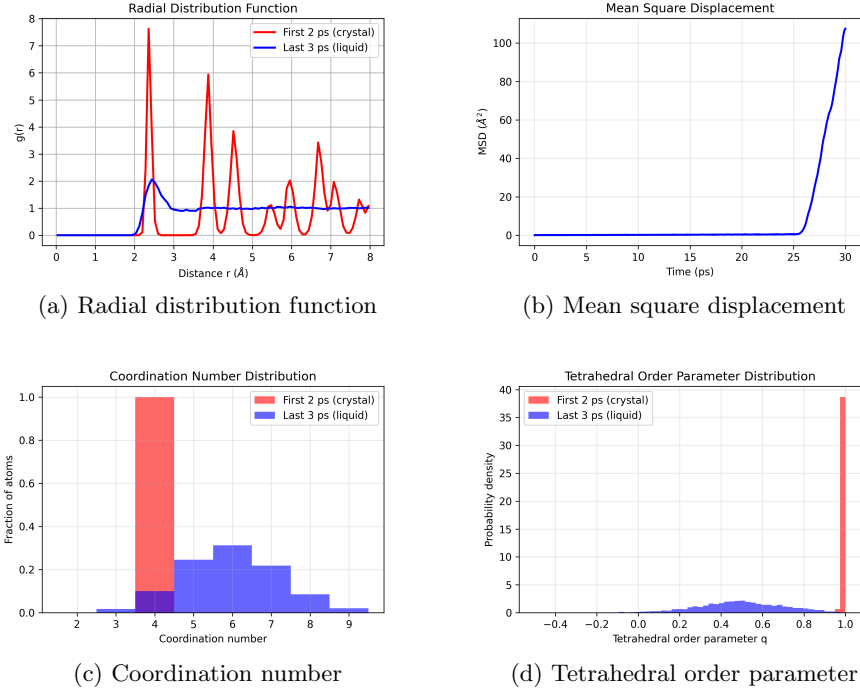


Figure 6: Melt with 26.almtp, same convention as Fig. 5.

The two potentials produce the same liquid. Every quantity in Table 10 agrees to better than 1.5%: the first peak of $g(r)$ to under one histogram bin, mean coordination to 0.07%, density to 0.34%, and the melting transition itself to one dump frame (25.6 against 25.7 ps). Rebuild counts, 400 against 374, are likewise indistinguishable. None of these differences is resolvable given the scatter of a 216-atom cell.

Potential	Steps	Wall time	Steps/s	Rebuilds	Neighs/atom
22.almtp	3.0×10^4	0:17:47	28.1	400	91.0
26.almtp	3.0×10^4	0:44:13	11.3	374	92.5

Table 11: Melt runs in 30 ps. No dangerous builds in either.

4.2 Hold

The melted configuration was held at 2500 K for 1 ns with each potential in turn, continuing directly from the saved state via `read_data` rather than rebuilding a lattice.

All five completed without instability. This is the first substantive result: the pruned potentials were expected to be unreliable at melt temperatures and

were not, showing no blow-up, no dangerous neighbour-list builds, and potential energies matching the level-22 reference to within 0.002%. The **26.almtp** hold was stopped at 400 ps, since at 10.4 steps/s a full nanosecond would have consumed the same wall-time budget for no additional information.

Run	Potential	Steps	Wall time	Steps/s	Rebuilds	Neighs/atom
Hold (1 ns)	22.almtp	1.0×10^6	10:35:32	26.2	30,215	94.8
Hold (0.4 ns)	26.almtp	4.0×10^5	10:42:19	10.4	12,134	93.6
Hold (1 ns)	pruned_1.almtp	1.0×10^6	0:54:24	307.2	30,200	91.2
Hold (1 ns)	pruned_2.almtp	1.0×10^6	2:30:32	110.7	30,164	94.8
Hold (1 ns)	pruned_3.almtp	1.0×10^6	6:31:03	42.6	30,243	94.5

Table 12: Hold runs at 2500 K on 8 MPI tasks ($\geq 98.7\%$ CPU use throughout). The **22.almtp** and pruned holds start from the **22.almtp** melt.

Computational cost. The cost ordering **pruned_1** < **pruned_2** < **pruned_3** < level 22 < level 26 comes out cleanly, with **pruned_1** and **pruned_2** running $11.7\times$ and $4.2\times$ faster than level 22 ($29.6\times$ and $10.7\times$ against level 26).

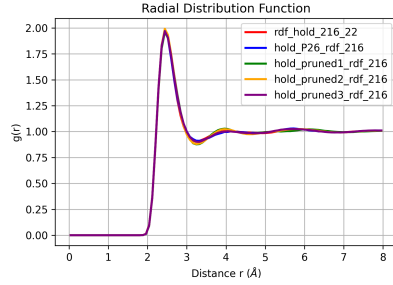
Potential 22 vs 26. Level 26 is $2.53\times$ slower than level 22 in the hold and $2.49\times$ slower in the melt. The same factor in both means the extra cost is per force call and does not depend on whether the system is a hot liquid or a cold crystal.

4.2.1 Structural Comparison

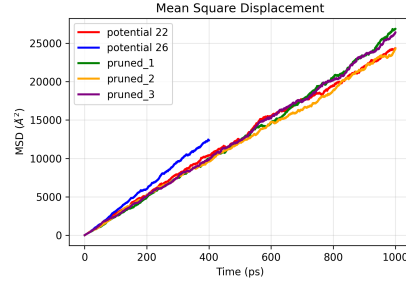
All five potentials give the same liquid. Wall times and energies say nothing about structure. The hold is isothermal, so there is no crystal-to-liquid contrast within a run, and the comparison is made across potentials with **22.almtp** as the reference, over the final 30 ps of each trajectory.

Potential	r_1 (Å)	$r_{\min}$ (Å)	$\bar{n}$	q	D (10^{-8} m ² /s)	ρ (g/cm ³)
22.almtp	2.44	3.24	5.960	0.4903	3.70	2.32
26.almtp	2.44	3.32	5.949	0.4821	5.26	2.31
pruned_1.almtp	2.44	3.24	5.925	0.4966	4.79	2.30
pruned_2.almtp	2.44	3.24	5.912	0.4975	4.08	2.29
pruned_3.almtp	2.44	3.32	5.970	0.4852	4.46	2.31

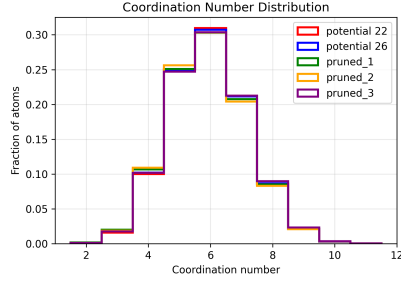
Table 13: The liquid at 2500 K for each potential, over the final 30 ps of each hold. r_1 and $r_{\min}$ are the first peak and first minimum of $g(r)$; the second peak is a shoulder of $g(r) \approx 1.02$ near 4 Å and too weak to locate reliably. $\bar{n}$ uses a 3.15 Å cutoff, D comes from the Einstein relation, ρ from the mean box volume. The **26.almtp** hold is 0.4 ns, so its dynamical statistics are correspondingly poorer.



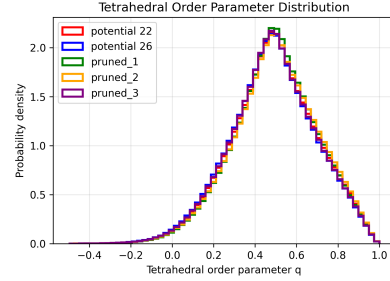
(a) Radial distribution function



(b) Mean square displacement



(c) Coordination number



(d) Tetrahedral order parameter

Figure 7: All five hold trajectories at 2500 K over the final 30 ps of each run. The $g(r)$, coordination and tetrahedral-order curves are indistinguishable between potentials. The MSD is linear and unbounded for every potential, confirming a fully diffusive liquid; 26.almtp is the only visible outlier and also the only 0.4 ns run.

Radial distribution function. The first peak of $g(r)$ falls at 2.44 Å for every one and the curves overlay within their linewidth.

Coordination number. Mean coordination runs from 5.912 to 5.970 across the five, a spread of 1.0%.

Tetrahedral order parameter. The tetrahedral order parameter ranges from 0.4821 to 0.4975, a spread of 3.1%.

Difference between pruned potentials. What matters more is that the leftover differences do not order by pruning level. `pruned_3` is the most accurate variant and gives the highest coordination of all five, above both melt-stable ones; `pruned_2` gives the lowest. If pruning were degrading the result the errors would grow steadily as descriptors are removed. They do not, so this is sampling noise.

Diffusion coefficient. The dynamics are much looser. D runs from 3.70 to $5.26 \times 10^{-8} \text{ m}^2/\text{s}$, a spread of 42%, but it comes from a single time origin and the `22.almtp` run is 0.4 ns against 1 ns for the others. Nothing should be read into it without a multiple-time-origin analysis.

Hand-off. The hand-off is justified rather than assumed. At $D \sim 4 \times 10^{-8} \text{ m}^2/\text{s}$ over 1 ns each atom covers of order 100 Å, several box lengths, so the liquid loses all memory of its starting configuration. The state passed to the quench does not depend on which potential ran the melt, and the $11.7\times$ saving costs nothing structurally in the liquid.

4.3 Quench

Each liquid was cooled from 2500 K to 300 K at 1 K/ps under NPT at 0 bar (2.2×10^6 steps), followed by a 1.5×10^5 -step isothermal equilibration at 300 K. All four holds continue from the `22.almtp` melt, and each potential quenched its own hold.

Potential	Total wall	Steps/s ³¹	T (K)	E_{pot} (eV)	V (Å ³)	ρ (g/cm ³)
<code>22.almtp</code>	24:34:31	29.2	315.8	−137,279.75	4482.4	2.2491
<code>pruned_1.almtp</code>	2:50:46	313.5	292.0	−137,276.65	4397.9	2.2881
<code>pruned_2.almtp</code>	6:31:05	106.9	316.8	−137,279.86	4477.9	2.2479
<code>pruned_3.almtp</code>	14:31:41	47.8	296.7	−137,276.83	4452.6	2.2613

Table 14: State variables are averaged over the last five printed steps of the 300 K equilibration; the scatter in T matches the ± 17 K expected from $\sqrt{2/3N}$ at $N = 216$. The `pruned_3` run was repeated after an earlier attempt was killed by a filesystem I/O fault at step 2.94×10^5 , and completed 5% under the 15.3 h projected from its hold throughput. No dangerous builds in any run.

Speed. The speedups are smaller here than in the hold. `pruned_1` gives $10.7\times$ over `22.almtp` against $11.7\times$ before, and `pruned_2` $3.7\times$ against $4.2\times$. `pruned_3` is unchanged at $1.6\times$.

Computing forces is not the only cost. Cores also spend time swapping atoms with each other, and that cost shrinks more slowly than the force cost, so its share of the run grows and the speedup falls short of the hold value.

Potential energy. Potential energies match to 0.015 eV/atom, comfortably inside $k_B T = 0.026$ eV at 300 K.

Density. `pruned_2` lands within 0.05% of `22.almtp`, `pruned_3` sits 0.5% above, and `pruned_1` 1.7% above.

³¹Measured over the 300 K block only, since LAMMPS reports one timing summary per run command.

Why pruned_1 is densest. The `pruned_1` result is the one predicted back in Section 3.3.2. In a fixed box where `22.almtp` sat at zero pressure, `pruned_1` held $-12,400$ bar, meaning it was pulling inward and wanted a smaller box. Here the box is free to move, so it takes one. Note that this flips between phases: `pruned_1` is among the least dense in the 2500 K liquid and the most dense at 300 K. Its 2.288 g/cm^3 is also the closest of the four to the experimental a-Si value of 2.28, with `22.almtp` 1.4% low at 2.249.

Expansion on freezing. All four expand on freezing, by 3.1%, 0.5%, 1.8% and 2.1%. That is the right direction for silicon and confirms the quench produced a solid rather than a stuck liquid. These come from the two-decimal liquid densities in Table 13, so the last digit is uncertain and the `pruned_1` figure is consistent with no change at all.

4.3.1 Filesystem Failure

The first `pruned_3` quench attempt terminated at step 2.94×10^5 with an I/O error on the dump write (`dump.cpp:543`). The error was EIO rather than EDQUOT and scratch stood at 351 MB of 20 TB, so the cause was a transient Lustre fault rather than a quota.

4.3.2 Structural Comparison

The quench produced amorphous silicon, not a frozen liquid and not a crystal. The clearest evidence is the second peak of $g(r)$: it survives at $3.88 \text{ \AA}$ in all four runs, where the liquid had nothing beyond $3.5 \text{ \AA}$. The third shell of the crystal at $4.50 \text{ \AA}$ is gone. Keeping the second peak while losing the third is the defining signature of a continuous random network.

Potential	r_1 ($\text{\AA}$)	$r_{\min}$ ($\text{\AA}$)	r_2 ($\text{\AA}$)	$\bar{n}$	defects (%)	q	MSD ($\text{\AA}^2$)	ρ (g/cm^3)
<code>22.almtp</code>	2.36	2.84	3.88	4.019	3.7	0.9304	0.050	2.2491
<code>pruned_1.almtp</code>	2.36	2.92	3.88	4.081	8.0	0.9205	0.055	2.2881
<code>pruned_2.almtp</code>	2.36	2.92	3.88	4.040	4.2	0.9302	0.050	2.2479
<code>pruned_3.almtp</code>	2.36	2.84	3.88	4.013	4.0	0.9286	0.047	2.2613

Table 15: The quenched solid at 300 K. r_1 , $r_{\min}$ and r_2 are the first peak, first minimum and second peak of $g(r)$. Coordination $\bar{n}$ uses a $2.85 \text{ \AA}$ cutoff, and “defects” is the fraction of atoms that are not fourfold. These columns are computed over the last 50 ps of each equilibration. The MSD column is the plateau value, averaged past 10 ps of the full 150 ps block, since the MSD is measured relative to the first frame of whatever window it is given. No diffusion coefficient is quoted: the MSD is flat, so D is not measurable.

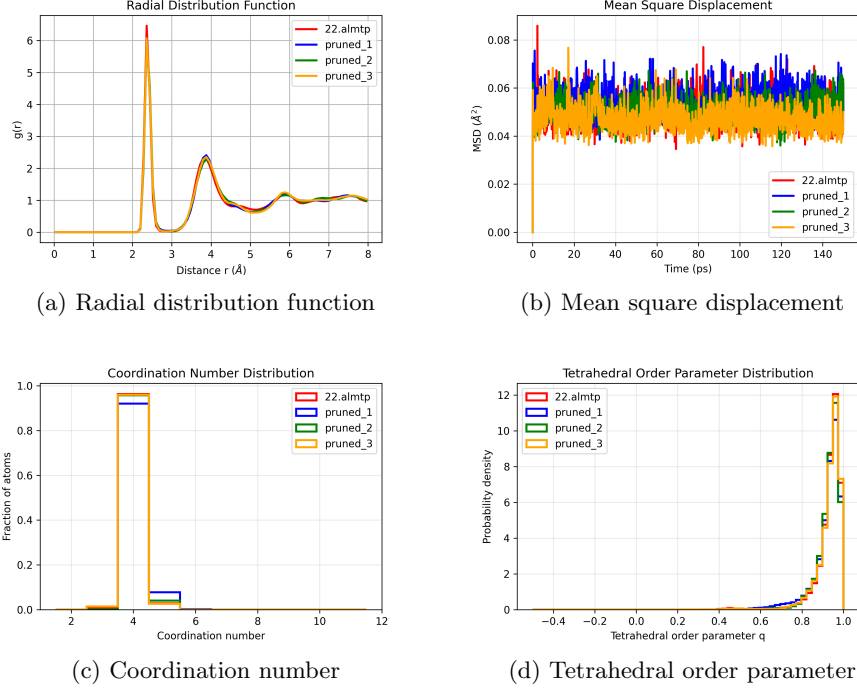


Figure 8: The quenched solid at 300 K, over the last 50 ps of each run, except the MSD which uses the full 150 ps. The MSD is flat for all four, confirming the atoms vibrate rather than diffuse. Coordination peaks sharply at fourfold and the tetrahedral order parameter near 0.93.

Coordination cutoff. The 2.85 Å cutoff taken from Zongo et al. lands almost exactly on the measured first minimum, 2.84 Å for two potentials and 2.92 Å for the other two, so the coordination counts do not depend on the choice.

Coordination number. Mean coordination is close to four for all four potentials, but the mean hides the interesting number. Between 3.7 and 8.0% of atoms are not fourfold, split between threefold dangling bonds and fivefold floating bonds. `pruned_1` is the worst at 8.0%, almost all of it fivefold, and it is also the densest run: packing the atoms closer forces extra neighbours inside the cutoff.

Tetrahedral order parameter. Tetrahedral order recovers to 0.93, from 0.49 in the liquid and against 0.99 for the crystal. The network has its tetrahedral geometry back but with the bond angles spread out, which is what a continuous random network is.

Mean square displacement. The MSD settles at about $0.05 \text{ \AA}^2$ and stays there for the full 150 ps. Fitting a slope gives $D \sim 10^{-14} \text{ m}^2/\text{s}$, six orders of magnitude below the liquid, and one of the four slopes comes out negative. The atoms are vibrating in place, not moving. This is a solid.

Comparison with Zongo et al. Zongo et al. [1] built two 216-atom models with the same potential family by melt-quench MD, reporting 1.85% and 1.86% defects. These runs sit two to four times higher. Part of the gap is method: their structures are relaxed to 0 K, while these are averaged over 500 frames at 300 K, where an atom vibrating near the $2.85 \text{ \AA}$ cutoff flickers in and out of the count.

Correlation length. Fitting $r |g(r) - 1| \sim e^{-r/\xi}$ through the extrema of $g(r) - 1$ between $3 \text{ \AA}$ and $L/2 = 8.3 \text{ \AA}$ does not give a usable number. Only six or seven extrema fall in that window, the four potentials return ξ between 1.26 and $2.89 \text{ \AA}$, and R^2 ranges from 0.40 to 0.60. The low values are unphysical: $\xi \approx 1.3 \text{ \AA}$ is shorter than a Si–Si bond and would rule out the second peak at $3.88 \text{ \AA}$ that all four runs show. The box is the limit. Measuring ξ properly needs one several times the expected 7 to $8 \text{ \AA}$.

Vibration amplitude. Atoms stay near their sites. The mean amplitude is 0.139 to $0.150 \text{ \AA}$ across the four potentials, rms 0.152 to $0.166 \text{ \AA}$, which matches the MSD plateau of Table 15.

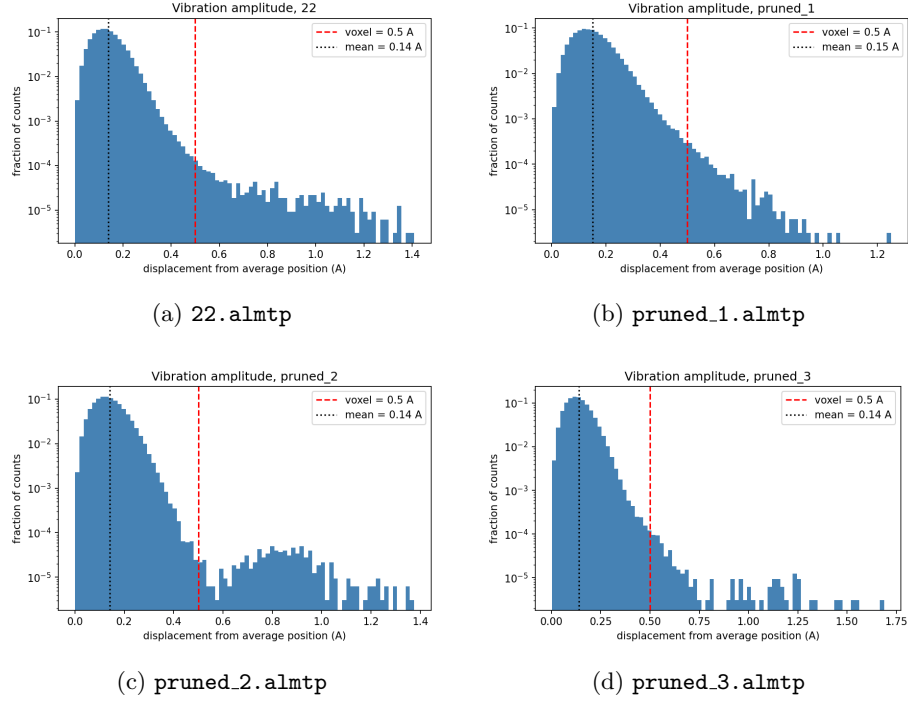


Figure 9: Distribution of atomic displacement from the average position, over the full 150 ps equilibration at 300 K. Coordinates are unwrapped and the centre of mass drift is removed. The dashed line marks the 0.5 Å voxel size.

Maximum displacement. No atom moves by its own size. The largest displacement from the start is 1.61, 1.47, 1.34 and 1.89 Å, all under the 2.35 Å bond length. Repeating over 1, 5 and 10 ps windows, which would catch a jump that reverses, reaches 1.79 Å at most. The spikes that appear come back down.

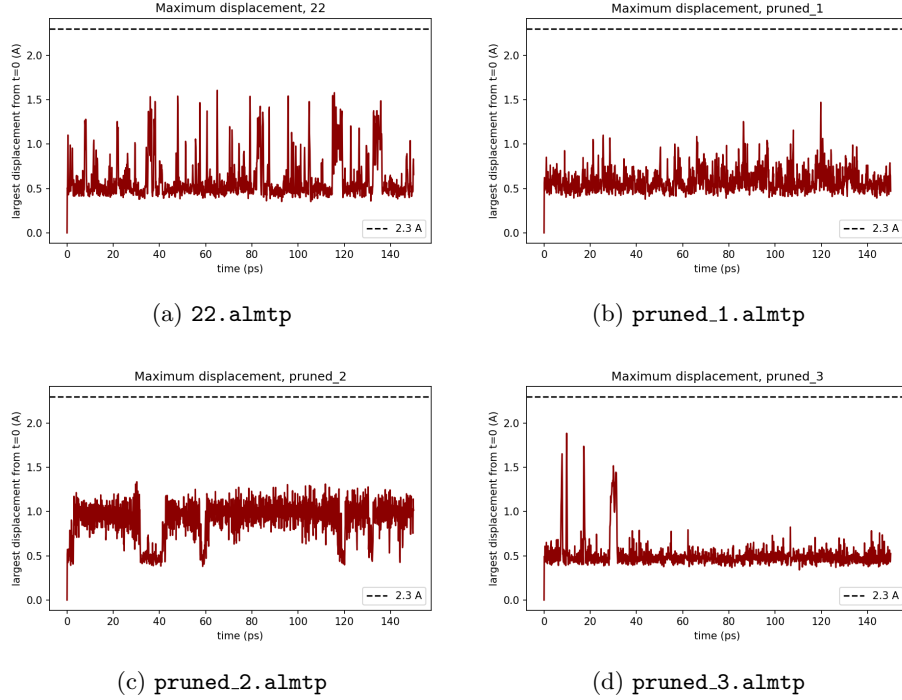


Figure 10: Largest displacement from the starting position at each frame, over the full 150 ps equilibration at 300 K. The dashed line marks 2.3 Å, roughly one Si-Si bond length. No potential reaches it.

Defect breakdown. Splitting the non-fourfold atoms by coordination shows that overcoordination dominates in every run. Floating bonds outnumber dangling bonds three to one in 22.almtp and two to one in pruned_3, while pruned_1 has no threefold atoms at all and is the only run to produce sixfold ones.

Potential	$\bar{n}$	3-fold (%)	4-fold (%)	5-fold (%)	6-fold (%)
22.almtp	4.019	0.9	96.3	2.8	—
pruned_1.almtp	4.081	—	92.0	7.8	0.2
pruned_2.almtp	4.040	0.1	95.8	4.1	—
pruned_3.almtp	4.013	1.4	96.0	2.6	—

Table 16: Coordination breakdown of the quenched solid at 300 K, using a 2.85 Å bond cutoff over the last 50 ps of each equilibration. Entries below 0.1% are omitted. pruned_1 is both the densest run (Table 15) and the most overcoordinated, consistent with closer packing forcing additional neighbours inside the cutoff.

Correlation between defects and large excursions. Coordination and displacement were recomputed as matched (frame, atom) arrays so each atom’s neighbour count could be read at the moment of its own displacement. For 22.almtp, 146 of 108,000 sampled points³² lie beyond the 0.5 Å voxel size, and 96.6% of those are non-fourfold against a 3.7% background. The effect is carried entirely by the undercoordinated atoms, which is the expected asymmetry: a dangling bond is missing a neighbour and is correspondingly less constrained. The tail is not a population effect, however. It comes from nine atoms, two of which contribute 128 of the 146 points.

	Overall ³³ (%)	In tail (%)	Enrichment
Non-fourfold	3.7	96.6	26×
3-fold	0.9	93.2	104×
5-fold	2.8	3.4	1.2×

Table 17: Coordination of the atoms in the > 0.5 Å tail of the displacement distribution for 22.almtp, over the last 50 ps at 300 K with a 2.85 Å cutoff. The tail holds 146 of 108,000 samples, so the percentages carry counting noise.

Crystallinity. The quenched structures were checked for leftover crystal order with the “identify diamond structure” modifier in OVITO, the same test used by Zongo et al. [1]. One frame is not enough: across the 1501 frames of the 22.almtp equilibration the count jumps between 17 flagged atoms and none, with nothing in between. This is a single diamond-like group (one atom, its four nearest neighbours and twelve second neighbours) sitting right on the edge of what the modifier counts, so it drops in and out as the atoms vibrate. It is counted in 56.4% of frames, which gives a time-averaged crystallinity of 4.5%. A hexagonal group shows up in 33 frames. Taking one frame alone would give either 7.9% or zero, depending on which.

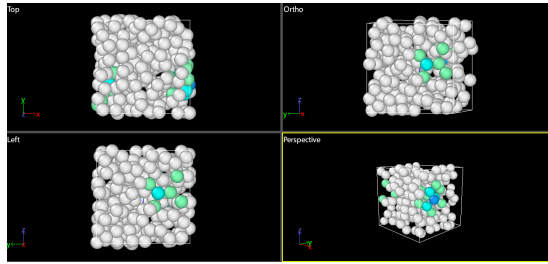


Figure 11: The quenched 216-atom cell for 22.almtp at 300 K, coloured by structure type, in a frame where the cubic motif is classified. White atoms are unclassified.

³²500 frames $\times$ 216 atoms = 108,000 cells. Each cell is one atom at one moment.

³³Overall of all 108,000 cells.

A Wooten-Winer-Weaire Algorithm

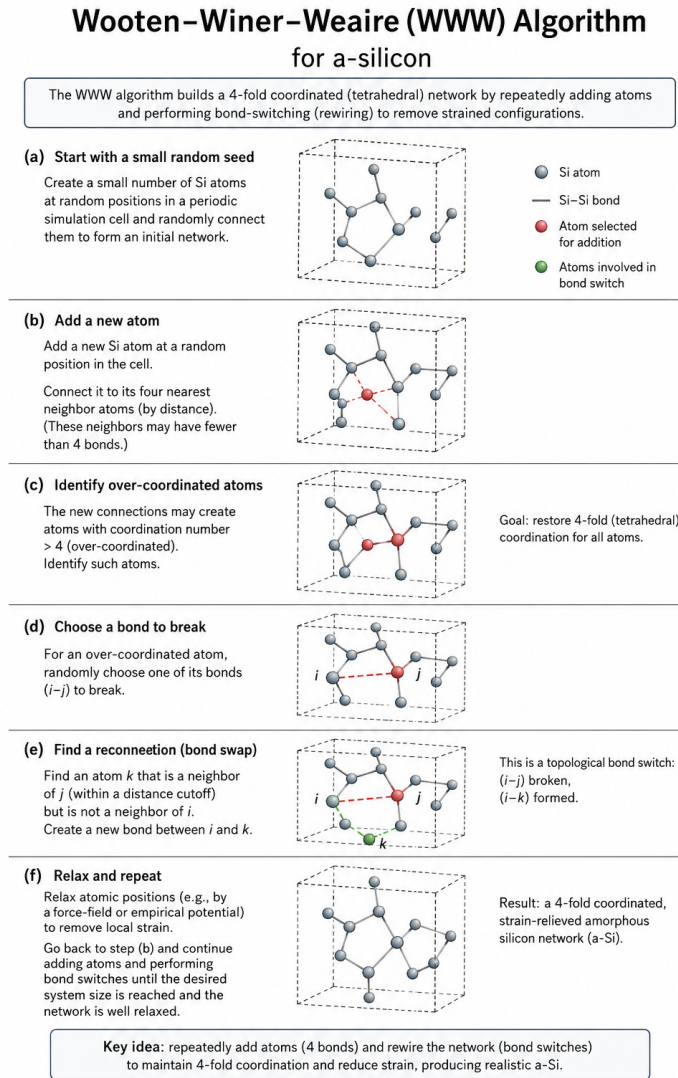


Figure 12: Schematic of the Wooten-Winer-Weaire (WWW) algorithm for generating continuous random network models of a-Si.

B A Plain-Language Note on DFT

DFT is a way to compute how electrons behave in a material without solving the full, extremely hard quantum equation directly. Normally, describing a

system of many electrons requires a huge mathematical object called a wavefunction that depends on the position of every electron at once, and this becomes impossible to handle as the number of electrons grows. DFT avoids this by working instead with the electron density $n(\mathbf{r})$, which simply describes how the electrons are spread out in space. Because it depends only on position in 3D, rather than on every electron individually, it is far cheaper to work with.

C Bond-Orientational Order Parameters

Definition 59 (Steinhardt Bond-Orientational Order Parameters). *The Steinhardt parameters q_ℓ score the shape of an atom's local neighbourhood. They discard how far the neighbours are and look only at the directions they sit in, returning a single number that says how symmetric that arrangement is.*

Point a unit vector $\vec{r}_{ij}$ at each neighbour j of atom i and average those directions against the spherical harmonics $Y_{\ell m}$:

$$q_{\ell m}(i) = \frac{\sum_j \sigma(r_{ij}) Y_{\ell m}(\vec{r}_{ij})}{\sum_j \sigma(r_{ij})}, \quad q_\ell(i) = \sqrt{\frac{4\pi}{2\ell+1} \sum_{m=-\ell}^{\ell} |q_{\ell m}(i)|^2}. \quad (11)$$

The $Y_{\ell m}$ are the functions that give atomic orbitals their shapes: ℓ fixes the lobe pattern on a sphere ($\ell = 0$ spherical, $\ell = 1$ the dumbbell, $\ell = 2$ the cloverleaf) and m labels the orientations of that pattern³⁴.

The weight $\sigma(r_{ij})$ sets how much each neighbour counts, with the denominator normalizing the average. The simplest choice is an indicator, 1 inside a cutoff and 0 outside, but an atom vibrating near the cutoff is then repeatedly counted and dropped, and q_ℓ jumps each time. Allowing σ to decay smoothly through intermediate values gives such atoms a fractional weight and removes the discontinuity, as does a Voronoi face-area weighting³⁵.

The second equation collapses the $2\ell + 1$ complex numbers $q_{\ell m}$ into one real number. Rotating the coordinate axes shuffles them among themselves but leaves the sum of their squares untouched, exactly as rotating a vector changes v_x, v_y, v_z while preserving $|\vec{v}|$. Since the orientation of the simulation box is arbitrary, only q_ℓ carries physical meaning.

The phases are then distinguished by cancellation. For $\ell \geq 1$ every lobe pattern averages to zero over the sphere, so it is positive on some regions and negative on others, much as $\sin x$ has humps above and below the axis. Randomly scattered neighbours therefore contribute with mixed signs and largely

³⁴A hydrogen wavefunction factorizes as $\psi = R_{n\ell}(r)Y_{\ell m}(\theta, \varphi)$: the radial part R describes how the amplitude falls off with distance, the angular part Y describes the shape. Steinhardt keeps only the angular part, which is why distance drops out. For each ℓ there are $2\ell + 1$ independent orientations, the same counting that gives three p orbitals and five d orbitals.

³⁵Space is divided so that every point belongs to its nearest atom. Two atoms count as neighbours when their cells share a face, and σ is taken as the area of that face: a large shared face means a strongly neighbouring atom, a sliver means a marginal one. This removes the cutoff parameter entirely [24].

cancel, whereas neighbours sitting at the special positions of a lattice can land on lobes of matching sign and reinforce. Computed directly from the ideal tetrahedral geometry, diamond-cubic c-Si gives $q_4 = 0.509$ and $q_6 = 0.628$; liquid silicon collapses toward zero with a broad spread and no sharp peak.

Reference values are tabulated for fcc, hcp, bcc, and icosahedral clusters [23]. No diamond entry appears there, and the omission is not an oversight: q_4 and q_6 cannot distinguish a tetrahedral environment from bcc, since the bcc motif is two point-symmetric tetrahedra and even- ℓ harmonics are blind to inversion. A second problem compounds this at small neighbour counts. Randomly oriented neighbours do not give $q_\ell = 0$ but roughly $1/\sqrt{N_b}$, since the cross terms cancel while the diagonal terms survive; with only four neighbours the random value is $q_4 \approx 0.49$, effectively indistinguishable from the ideal tetrahedral 0.509. Tetrahedral silicon is therefore the worst case for these parameters, which is why the tetrahedral order parameter of Section 2.5.4 is used in this project instead.

D Work Log

Date	Activity
July 13	CPU build of LAMMPS-MTP debugged and validated. Three faults in the supplied archive resolved: incompatible architecture flags, an inverted conditional that skipped the source copy, and stale nested CMake caches. Lennard-Jones validation run completed in 4 min 16 s.
July 14	GPU (Kokkos) build debugged and validated. Same Lennard-Jones case ran in 15.6 s on an A100, a speedup of roughly 16 $\times$.
July 21	Atom type mapping settled by two NVT tests: type 1 is silicon, type 2 oxygen.
July 22	Parallel scaling measured on 1, 4 and 8 cores. Eight cores adopted as the default for 216 atoms.
July 23	Cooling-rate protocol fixed. Zongo’s 1 K/ps from 2500 to 300 K sets the quench at 2.2×10^6 steps.
July 24	Melt of the 216-atom diamond lattice to 2500 K with <code>22.almtp</code> . Melting confirmed on five independent signatures.
July 27	Interim report submitted.
July 28	1 ns hold at 2500 K with <code>22.almtp</code> completed in 10 h 35 min. Holds repeated with <code>pruned_1</code> and <code>pruned_2</code> .
July 29	Hold with <code>pruned_3</code> completed in 6 h 31 min, establishing that all three pruned potentials survive a nanosecond in the liquid.
July 30	Melt and 0.4 ns hold with <code>26.almtp</code> .

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Date	Activity
August 1	The pruned_3 quench terminated at step 2.94×10^5 by a filesystem I/O fault (Section 4.3.1).
August 3	Structural comparison of all five hold trajectories. Quench results collated for 22.almtp , pruned_1 and pruned_2 .
August 5	All four quenches complete and analysed. Amorphous silicon confirmed: second peak of $g(r)$ at 3.88 Å with the third absent, flat MSD, tetrahedral order recovered to 0.93. Defect fractions of 3.7–8.0% against Zongo’s 1.85%.
August 7	Correlation length fitted from $g(r)$. The 216-atom box limits the fit to 8.3 Å and returns unphysical values, so the measurement needs a larger cell.
August 10	Vibration amplitudes and maximum displacements computed for all four potentials. No atom moves by its own size over 150 ps; the largest excursion is 1.89 Å against a 2.35 Å bond length. Short-window analysis at 1, 5 and 10 ps finds no reversed jumps either.
August 11	Amplitude distribution replotted normalised and on a log axis at Ata’s request, revealing a flat tail from 0.5 to 1.2 Å above the ordinary vibrational peak.
August 17	MAP installed on Narval (~/MAP-main , virtual environment at ~/envMAP). Coordination defects broken down by type: overcoordination dominates in all four potentials, and pruned_1 has no threefold atoms at all. Correlation established between the > 0.5 Å tail and undercoordination. Crystallinity measured in OVITO across all 1501 frames, giving a time-averaged 4.5% carried by a single flickering motif.
August 19	MAP validation run on the bundled water data killed by the out-of-memory killer. Diagnosed as host RAM rather than GPU memory and resolved with --mem-per-gpu=100G , matching the configuration used by the MAP authors. The ckpts/ prefix was stripped from save_ckpt so checkpoints write to the run directory rather than the quota-limited home filesystem.

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Date	Activity
September 5	Water validation run completed. Training converged over 81 epochs, the test negative log-likelihood falling from 0.0102 to 0.0069 and flattening past epoch 58, at roughly 940 s per training epoch and 80 s per test pass. Generation of a single sample ran the full 80 steps in 1 h 17 min, then failed at the final write: <code>np.save</code> was called on the relative path <code>samples/run_0.npy</code> and no such directory existed. The same class of fault as the <code>ckpts/</code> prefix, and the second time MAP’s hardcoded relative output paths have cost a run.
September 14	Save path in <code>utils.py</code> redirected to <code>/scratch</code> , and the water run resubmitted in full. No checkpoint was written for the completed training, so all 81 epochs are repeated to reach the generation stage.

Table 18: Chronological record of the work described in Sections 3 and 4. All dates are 2026.

E Command References

E.1 PuTTY

E.1.1 Navigating and editing files (Linux)

```
cd /path/desired # Move into your desired folder.
ls # List files in the current folder.
ls -l # List files with details (size, permissions, owner, etc.).
ls -a # List all files, including hidden ones.
mkdir <foldername> # Create a new folder.
nano <filename> # Open a file in a simple text editor.
vim <filename> # Open a file in the vim text editor.
rm -rf <foldername> # Delete a folder and all its contents, without asking
                    # for confirmation. No undo, check the path first.
rm -f <filename> # Delete a file.
cp -r <source> <destination> # Copy a folder and its contents.
cat <filename> # Print a file’s contents to the terminal.
head -5 <filename> / tail -30 <filename> # Show the first 5 / last 30
                    # lines of a file.
grep -in "error" <filename> # Search a file for a word (case-insensitive)
                    # and show matching lines with line numbers.
cat > <filename> << 'EOF' ... EOF # Write everything between the mark-
                    # ers into a file (creates or overwrites it).
bash <scriptname> 2>&1 | tee <logname> # Starts a new bash shell session
```

to read and execute the commands in a script file. Merges errors into normal output. Shows output on screen and save it to a log file you name.

`mv <filename> <new_filename> #` Rename a file.
`diskusage_report #` Shows your current disk usage and quota. Accepts `--scratch`, `--home`, `--project` and `--nearline` to restrict the output to one filesystem. Both a size quota and a file-count quota are reported, and either can halt a running job.

E.1.2 Build Environment Commands

`module load gcc openmpi flexiblas cmake fftw eigen voro++ cuda #` Activates the software environment needed for compilation: `gcc` (the C/C++ compiler), `openmpi` (the MPI parallel communication library), `flexiblas` (linear algebra), `cmake` (the build-configuration tool), `fftw` (fast Fourier transforms), `eigen` (matrix mathematics), `voro++` (Voronoi analysis), and `cuda` (the NVIDIA GPU toolkit, needed only for the GPU build). On Narval, none of these are available until explicitly loaded.
`bash ./build-lmp-mtp-gpu.sh #` Executes the build script in a bash shell. The script clones the MTP repository if absent, copies the MTP source files into the LAMMPS source tree, configures the build with CMake, compiles, and installs the executable.

E.1.3 Job Submission Commands

`sbatch <script.sh> #` Submits a batch script to the queue.³⁶
`salloc --account=ctb-simine --gres=gpu:1 --cpus-per-task=8 --mem=16G --time=1:00:00`
Example of a simple interactive allocation for GPU build.
`salloc --account=ctb-simine --cpus-per-task=8 --mem=16G --time=1:00:00`
Example of a simple interactive allocation for CPU build.
`scancel <job_id> #` Cancels a pending or running job.
`tail slurm-<job_id>.out #` Shows the last 10 lines of the job's output file, useful for checking progress or errors.

For the full list of SLURM commands used to submit, monitor, and cancel jobs, see the accompanying reference sheet `SlurmCheatSheet.pdf`, distributed alongside this report.

With `salloc`, resource flags are typed directly on the command line because there is no script. With `sbatch`, the same flags are written inside the script as `#SBATCH` lines, so the command is simply `sbatch <script.sh>` with no flags needed.

Note that a GPU functions as a co-processor. A host CPU always runs the main program and delegates the computationally intensive kernels to the GPU, which is why GPU allocations request both CPU cores and a GPU.

³⁶Note that there should be different `script.sh` for GPU and CPU builds.

E.1.4 CMake Options Used in the Build Scripts

```
--fresh # Discards the top-level CMake cache so the configuration restarts
         from scratch. Note: nested sub-project caches are NOT reset, which is why
         deleting _build entirely was required.
-DPKG_ML-MTP=ON # Compiles the MTP pair styles into LAMMPS.
-DPKG_KOKKOS=ON # Compiles the Kokkos package (portable CPU/GPU parallelism).
-DKokkos_ENABLE_CUDA=ON # Enables the NVIDIA GPU backend of Kokkos.
-DKokkos_ARCH_AMPERE80=ON # Optimizes the GPU code for A100 (Ampere) GPUs.
-DCMAKE_CXX_FLAGS="-march=core-avx2" # Compiles CPU code for the AVX2 instruction set matching Narval's processors (CPU build only; incompatible with the CUDA compiler).
```

E.1.5 Running LAMMPS

```
lmp -in <input_file.in> # Run LAMMPS with a given input script.
lmp_mpi -in <input_file.melt> # Run the MPI (parallel) LAMMPS build.
```

E.2 Tk Console

E.2.1 Navigating to the files

```
cd /path/desired # Move into your desired folder.
ls # List files in the current folder.
```

E.2.2 Loading a single configuration (data file)

```
topo readlammpsdata <config.end_sim> atomic # Reads a LAMMPS data file written with atom_style atomic and loads it as a new molecule
```

E.2.3 Loading a full trajectory (many frames)

```
mol new <dump.lammpstrj> type lammpstrj waitfor all # Loads a LAMMPS trajectory, importing every frame so the run can be animated
```

E.2.4 Drawing the simulation box

```
pbc box # Draws the periodic simulation cell around the atoms
```

E.3 Input Script

LAMMPS reads an input script line by line, in order, so commands must appear before anything that depends on them: the unit system and lattice are defined first, then the box is created, then atoms are placed, then the potential and settings are assigned, and only then is **run** issued. Anything set after **run**

is ignored for that run. Blank lines and lines beginning with `#` are ignored, and the file extension (`.in`, `.melt`, `.lmp`) is only a naming convention.

units `<unit_system>` `#` Sets the unit system (either 'lj' or 'metal'). With 'lj', all quantities are dimensionless and expressed in terms of the Lennard-Jones parameters σ and ϵ . With 'metal', quantities carry physical units: distances in Angstroms, energies in eV, temperatures in kelvin, and time in picoseconds.

atom_style **atomic** `#` Specifies that each atom carries only a position, velocity, and type, with no bonds or charges. This is the simplest atom style and is appropriate for LJ fluids and metals.

lattice `<lattice_type>` `<density>` `#` Defines a lattice type (could be face-centred-cubic lattice by typing 'fcc' or diamond lattice by typing 'diamond') at reduced number density (i.e. '0.8442'). This sets the scaffold of points on which atoms will later be placed, but creates no atoms itself.

dimension **3** `#` Sets the dimensionality of the simulation to 3D.

boundary **p p p** `#` Sets periodic boundary conditions in all three dimensions.

read_data `<filename>` `#` Reads a LAMMPS data file containing a complete atomic configuration, including box size, atom positions, and velocities. This is an alternative to creating a lattice and populating it with atoms.

read_restart `<filename>` `#` Reads a binary restart file written by **restart** and resumes a run from the exact state it captured, including fix internals that **read_data** does not preserve. Note that a temperature ramp is defined relative to the **run** command that follows, so a resumed quench needs its start temperature and remaining step count recomputed by hand.

region **box** **block** **0 15 0 15 0 15** **units** **lattice** `#` Defines a cubic region named **box**, spanning 0 to 15 lattice units along each axis.

create_box **1** **box** `#` Creates the simulation box from that region, with space for a single atom type.

create_atoms **1** **region** `<region_name>` `#` Populates the named region (e.g. 'box') with type-1 atoms on the lattice points defined earlier. For example, a 15^3 fcc region with four atoms per unit cell yields 13,500 atoms, while a 3^3 diamond region with eight atoms per unit cell yields 216.

mass **1** **1.0** `#` Assigns a reduced mass of 1.0 to atom type 1.

velocity **all** **create** `!temperature` `!seed` `!loop` **geom** `#` Assigns random initial velocities to all atoms corresponding to the specified temperature (dimensionless in 'lj' units, kelvin in 'metal' units). The integer seed makes the run reproducible. 'loop geom' ensures the total momentum is zero, so the system does not drift.

pair_style lj/cut 2.5 # Selects the Lennard-Jones potential with a cutoff of 2.5 (reduced units); interactions beyond this distance are neglected to reduce cost.

pair_style mtp ;model_file; # Selects the Moment Tensor Potential, loading the trained model from the specified file. This is used for silicon simulations.

pair_coeff 1 1 1.0 1.0 2.5 # Sets the interaction parameters between type-1 atoms: $\epsilon = 1.0$, $\sigma = 1.0$, and cutoff 2.5.

neighbor 0.3 bin # When LAMMPS looks up which atoms are close enough to interact, it keeps a list of nearby neighbours. This adds a small 0.3 buffer (a “skin”) around each atom’s cutoff so that atoms drifting slightly closer are already on the list. Without the buffer, the list would have to be rebuilt every single step, which is slow.

neigh_modify delay 5 every 1 # Controls neighbour-list rebuild frequency: wait at least 5 steps, then check every step.

compute myRDF all rdf <hist.bins> 1 1 1 2 # Calculates the radial distribution function $g(r)$ and coordination number for a group of atoms. With hist.bins (i.e. 100) used to split the distance between 0.0 and the cutoff distance. Higher values yield smoother curves but require more memory. Whilst ‘1 1’ and ‘1 2’ Computes the $g(r)$ between type 1 atoms and type 1 atoms and between type 1 and type 2 respectively.

fix <ID> <group> <integrator> # Applies a time-integration scheme to a group of atoms. The first argument is a user-chosen ID naming this fix (commonly just ‘1’); the second is the group of atoms it applies to (‘all’ means every atom in the simulation). The integrator is either ‘nve’ (constant number, volume, and energy), ‘nvt’ (constant number, volume, and temperature), or ‘npt’ (constant number, pressure, and temperature, where the box is allowed to change size).

fix <ID> <group> ave/time <timesteps> <num_of_samples> <Avg_calculations_timesteps> c_myRDF[*]
Samples c_myRDF[*] (the full global array from compute rdf, with [*] pulling every column) every 10 timesteps, accumulates 100 such samples per block (1000 timesteps), treats each input as a histogram-bin vector via mode vector, and writes the resulting average to file every 1000 timesteps.

fix <ID> <group> npt temp <T.start> <T.stop> <T.damp> iso <P.start> <P.stop> <P.damp>
Runs the NPT ensemble with concrete values. The three numbers after ‘temp’ are the starting temperature, the target temperature, and the damping constant³⁷. The three after ‘iso’³⁸ are the same for pressure.

³⁷The damping constant controls how quickly the thermostat pulls the system back to the target, in picoseconds.

³⁸The box is rescaled equally in all three directions.

When the start and stop temperatures differ, as in `temp 300.0 2500.0 0.1`, LAMMPS ramps linearly between them over the run, which is how heating and cooling rates are set.

`unfix <ID> #` Removes a previously defined fix, identified by the ID given when it was created. Necessary when one stage of a run must be replaced by another, since fixes otherwise remain active for every subsequent `run` command. In a quench, the ramping thermostat is removed with `unfix` before a second `fix npt` holds the system at the final temperature.

`thermo_style custom step temp press vol pe #` Chooses which quantities are printed in the thermodynamic output. ‘custom’ means the list is user-specified: here the timestep number, temperature, pressure, box volume, and potential energy. Tracking ‘vol’ is essential in NPT runs, where the box size changes and must be monitored for convergence.

`thermo 5 #` Prints thermodynamic output³⁹ every 5 timesteps.

`dump 1 all atom 100 dump.lammpstrj #` Writes the atomic positions to a trajectory file every 100 timesteps, in a format readable by VMD.

`run 10000 #` Runs the simulation for 10,000 timesteps.

`restart <steps> <file1> <file2> #` writes a binary snapshot of the full simulation state (positions, velocities, box, fix internals) every `<steps>` steps, alternating between the two filenames so the latest and one fall-back are always on disk. Not for analysis; its only use is `read_restart` to resume a run killed before completion.

`write_data ;filename; #` Writes the final atomic configuration to a file for later analysis or restart.

E.4 Job Submission Script

`-account=ctb-simine #` Charges the job to the research group’s compute allocation.

`-ntasks=1 #` Requests a single processor core.

`-mem=4G #` Requests 4 GB of memory for the job.

`-time=0:15:00 #` Sets a wall-clock limit of 15 minutes, after which the job is terminated.

`module load ... #` Loads the required software environment: the compiler, MPI library, and the specific LAMMPS build.

`export PATH=... #` Points the shell to the custom-built LAMMPS executable instead of the system-wide module version.

³⁹Thermodynamic output includes properties like temperature, pressure, and energy.

lmp -in test.in # Executes LAMMPS using the prepared input script.

lmp -k on g 1 -sf kk -pk kokkos newton on neigh half -in test.in # Executes the GPU build of LAMMPS with Kokkos and MTP support, specifying GPU usage and neighbour-list options.

Example job submission script for the Lennard-Jones melt test using the CPU build of LAMMPS with MTP support:

```
#!/bin/bash
#SBATCH --account=ctb-simine
#SBATCH --time=0:15:00
#SBATCH --ntasks=1
#SBATCH --mem=4G

module load gcc openmpi flexiblas fftw eigen voro++
export PATH=$HOME/software/lammps-mtp-cpu/bin:$PATH

lmp -in test.in
```

Example job submission script for the same test using the GPU build. Note the four differences from the CPU script: the **--gres=gpu:1** line requesting a GPU, the **cuda** module added to the load line, the **PATH** pointing to the GPU build (**lammps-mtp-gpu**), and the Kokkos flags (**-k on g 1 -sf kk ...**) on the **lmp** command that activate GPU execution:

```
#!/bin/bash
#SBATCH --account=ctb-simine
#SBATCH --time=0:15:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --gres=gpu:1
#SBATCH --mem=8G

module load gcc openmpi flexiblas fftw eigen voro++ cuda
export PATH=$HOME/software/lammps-mtp-gpu/bin:$PATH

lmp -k on g 1 -sf kk -pk kokkos newton on neigh half -in test.in
```

Lines beginning with **#SBATCH** are resource requests to the scheduler, not commands.

The ‘**cuda**’ module loads the NVIDIA GPU toolkit, required for the GPU build.

F Further Viewing and References

The following items provide supplementary background for this report:

- Exciting-World. “Science — Molecular Dynamics Simulation — The New Class of Moment Tensor Potentials.” *YouTube*, 18 May 2022. <https://www.youtube.com/watch?v=dg5-FfXiSp8>.
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